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Singla et al.

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(54) **PREDICTIVE ADJUSTMENT OF
MULTI-CAMERA SURVEILLANCE VIDEO
DATA CAPTURE USING GRAPH MAPS**

(71) Applicant: **Sandisk Technologies, Inc.**, Milpitas,
CA (US)

(72) Inventors: **Lovish Singla**, Bangalore (IN);
Shaheed Nehal A, Mysore (IN)

(73) Assignee: **Sandisk Technologies, Inc.**, Milpitas,
CA (US)

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(56) **References Cited**

U.S. PATENT DOCUMENTS

7,583,815	B2	9/2009	Zhang et al.
7,697,827	B2	4/2010	Konicek
8,824,879	B2	9/2014	Konicek
9,031,847	B2	5/2015	Sarin
9,560,323	B2	1/2017	Huang
10,230,884	B2	3/2019	Okada
10,643,472	B2	5/2020	Song
10,701,321	B2	6/2020	Westmacott
10,855,921	B2	12/2020	Milne
10,885,342	B1	1/2021	Day
10,897,627	B2	1/2021	Muthiah
11,538,316	B2	12/2022	Lee
11,619,991	B2	4/2023	Kelly

(Continued)

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FOREIGN PATENT DOCUMENTS

CN	109300471	A	2/2019
JP	2015104028	A	6/2015

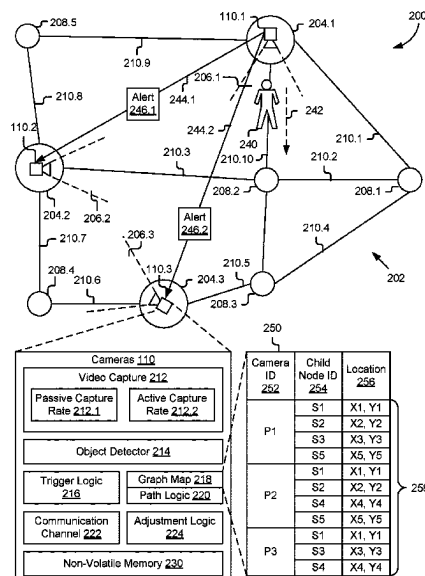
Primary Examiner — Rebecca A Volentine

(74) *Attorney, Agent, or Firm* — Patent Law Works LLP

(57) **ABSTRACT**

Systems, video cameras, and methods for predictive adjustment of multi-camera surveillance video data capture based on graph maps are described. A plurality of networked video camera is deployed and represented in a graph map based on the video camera environment, with parent nodes corresponding to video cameras and child nodes corresponding to path intersections among the video cameras. When a video event is detected from video data for one of the video cameras, a video capture update message indicating a shared child node identifier is selectively sent to other video cameras to modify their video capture operations.

20 Claims, 8 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2004/0257444	A1	12/2004	Maruya	
2005/0102133	A1	5/2005	Rees	
2005/0128311	A1	6/2005	Rees	
2007/0236570	A1	10/2007	Sun	
2008/0198231	A1	8/2008	Ozdemir	
2009/0245573	A1	10/2009	Saptharishi	
2009/0309973	A1	12/2009	Kogane	
2010/0141767	A1	6/2010	Mohanty	
2010/0157064	A1	6/2010	Cheng	
2013/0036628	A1	2/2013	Hussey	
2013/0208113	A1	8/2013	Yoneji	
2013/0314543	A1	11/2013	Sutter	
2014/0028826	A1	1/2014	Lee	
2014/0160316	A1	6/2014	Hwang	
2014/0192211	A1	7/2014	Konicek	
2014/0333775	A1	11/2014	Naikal	
2015/0172604	A1	6/2015	Kim	
2016/0189500	A1	6/2016	Kim	
2017/0127302	A1	5/2017	Fersman	
2017/0221219	A1	8/2017	Cho	
2018/0007323	A1	1/2018	Botusescu	
2018/0013658	A1 *	1/2018	Wiesmaier	H04L 41/12
2018/0165520	A1	6/2018	Meisser	
2019/0037178	A1	1/2019	Larsen et al.	
2019/0080575	A1	3/2019	Lee	

2019/0104283	A1	4/2019	Wakeyama	
2019/0156849	A1	5/2019	Feng	
2019/0258865	A1	8/2019	Ernesti	
2020/0005615	A1	1/2020	Madden	
2020/0105111	A1	4/2020	Messer	
2020/0234057	A1	7/2020	Koo	
2020/0327347	A1	10/2020	Arai	
2021/0201933	A1	7/2021	Kang	
2021/0289168	A1	9/2021	Glückert	
2021/0342619	A1	11/2021	Faller	
2021/0374971	A1 *	12/2021	Zhou	G06F 18/22
2021/0409790	A1	12/2021	Larrew	
2022/0171970	A1	6/2022	Yadav et al.	
2022/0215727	A1	7/2022	Jeong	
2022/0292833	A1	9/2022	Björn	
2022/0383522	A1	12/2022	Lai	
2022/0391090	A1	12/2022	Muthiah	
2022/0406065	A1	12/2022	Kang	
2023/0031871	A1	2/2023	Shroff	
2023/0064558	A1	3/2023	Griffin	
2023/0082600	A1	3/2023	Nakazato	
2023/0156337	A1	5/2023	Oh	
2023/0209204	A1	6/2023	Zu	
2023/0252654	A1	8/2023	Kamiya	
2024/0007744	A1	1/2024	Muthiah	
2024/0155221	A1	5/2024	Watanabe	
2024/0386581	A1 *	11/2024	Kamada	G06T 7/292

* cited by examiner

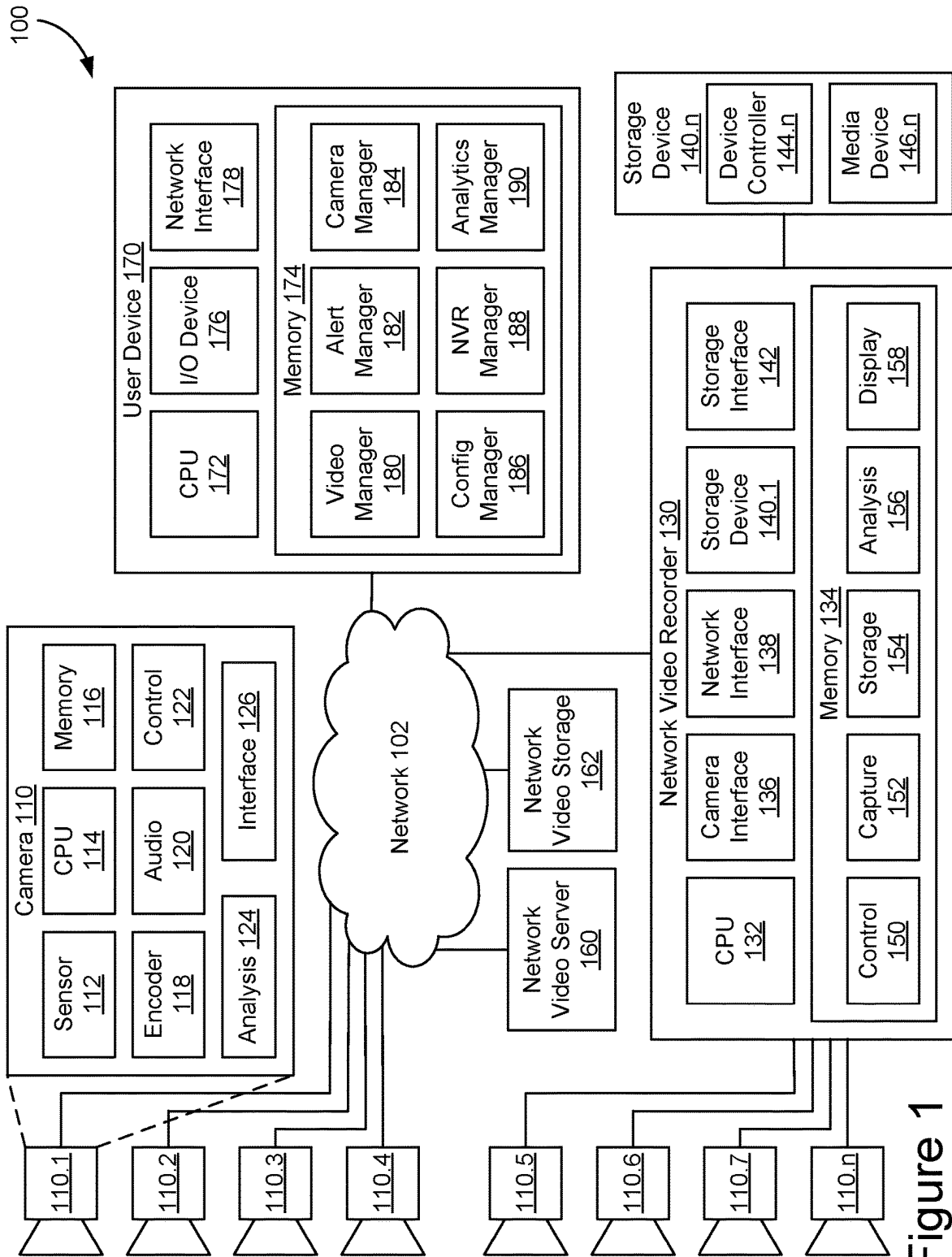


Figure 1

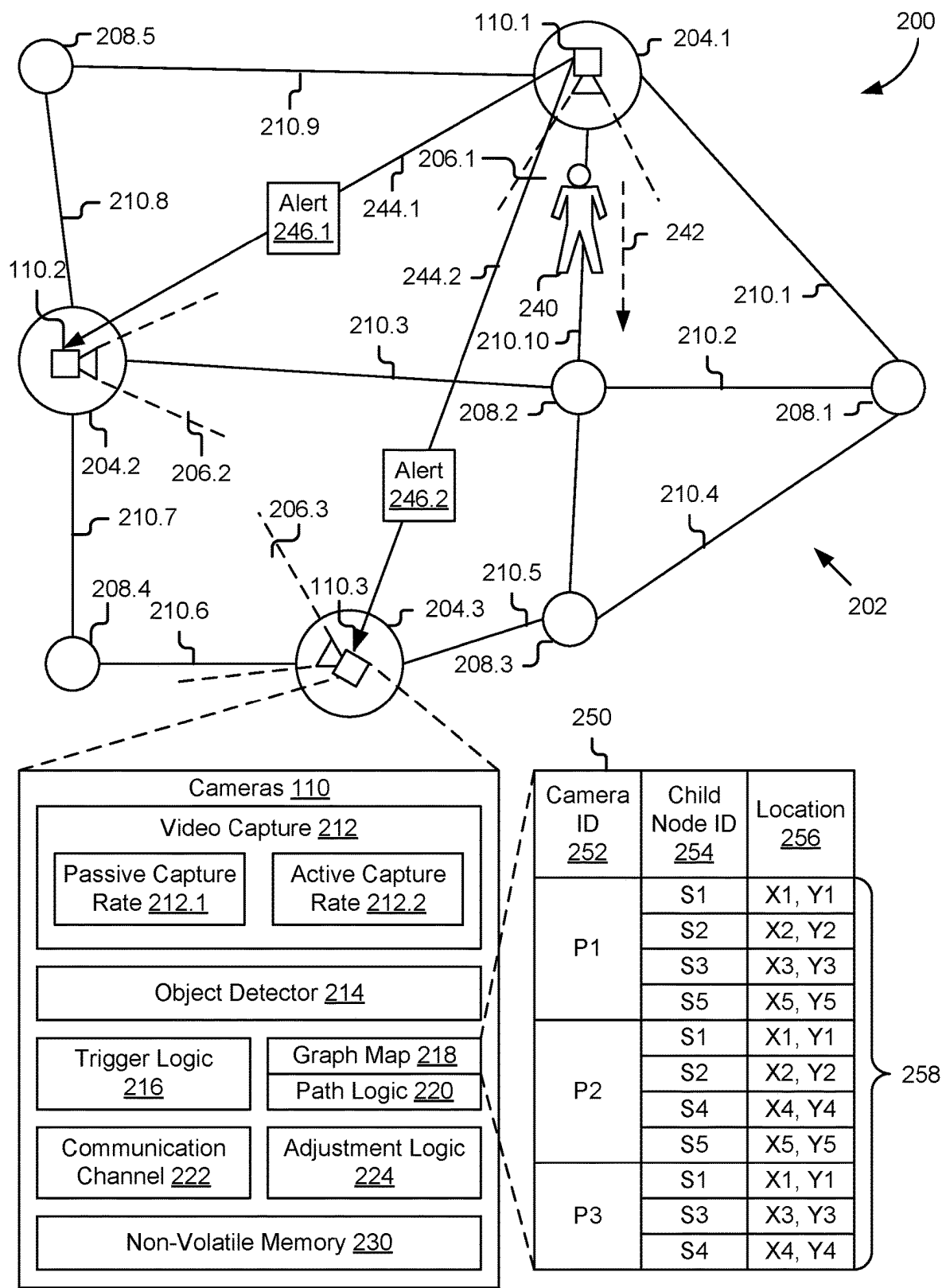
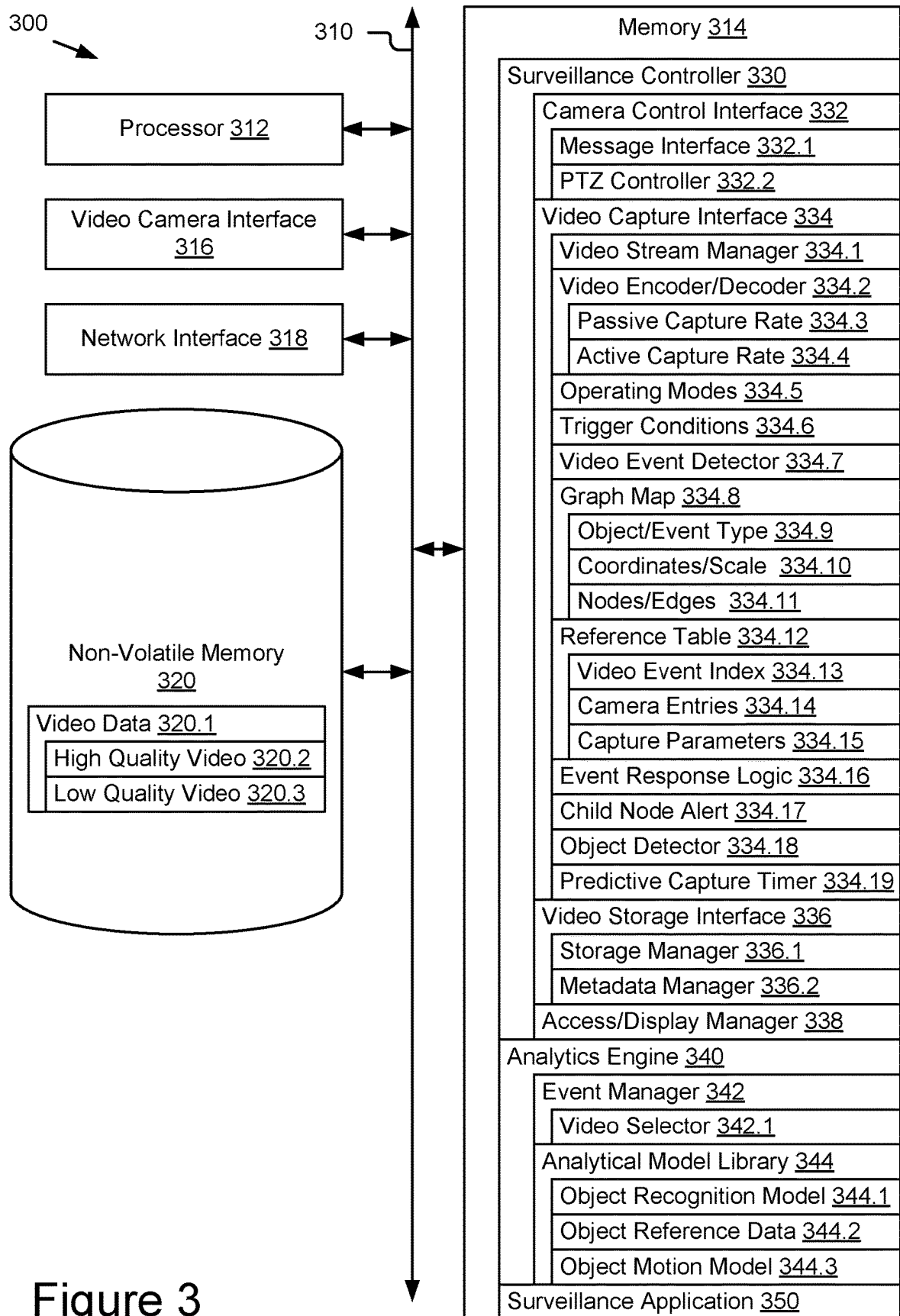


Figure 2



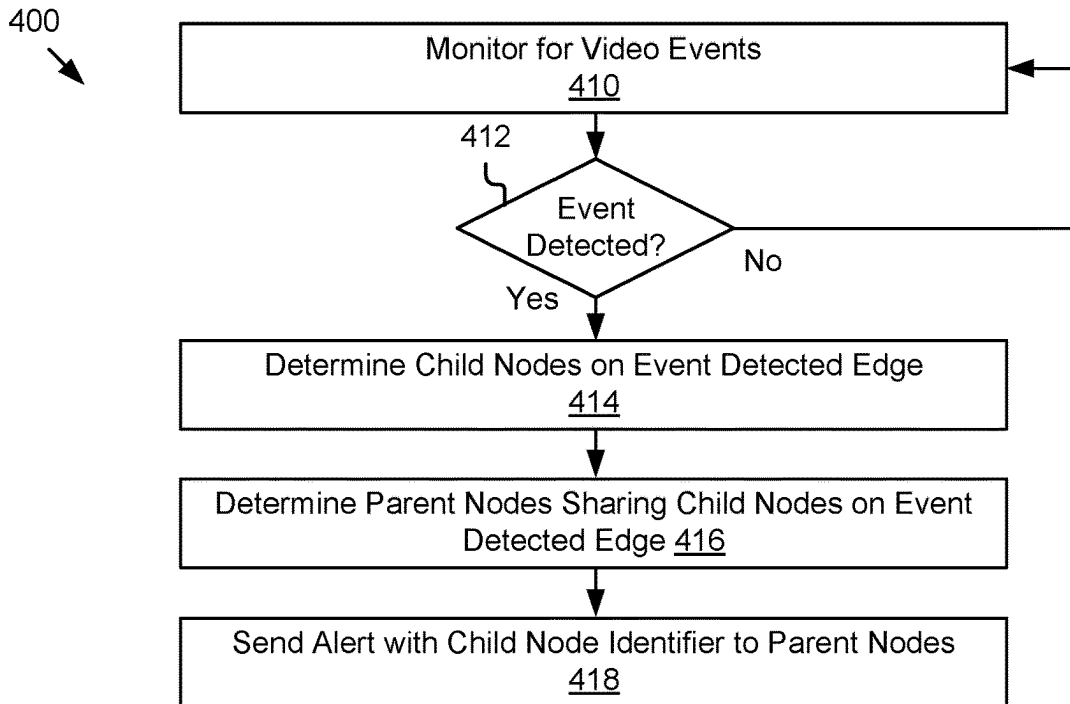


Figure 4

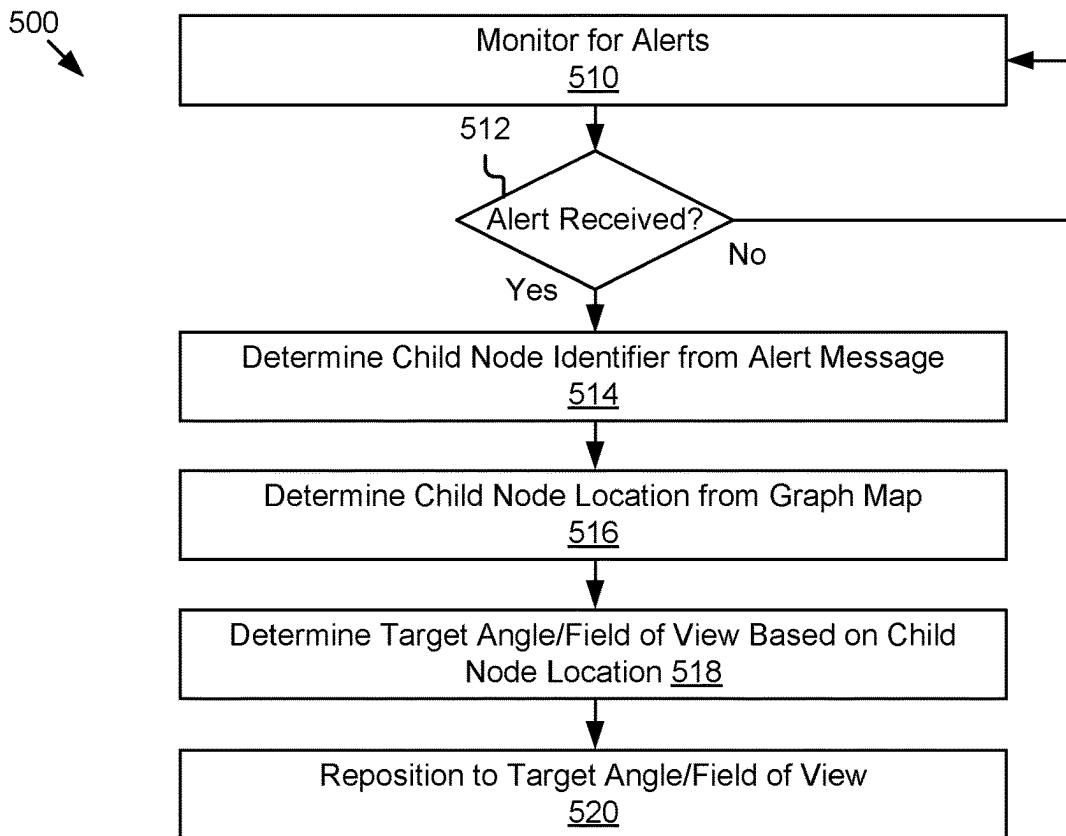


Figure 5

600

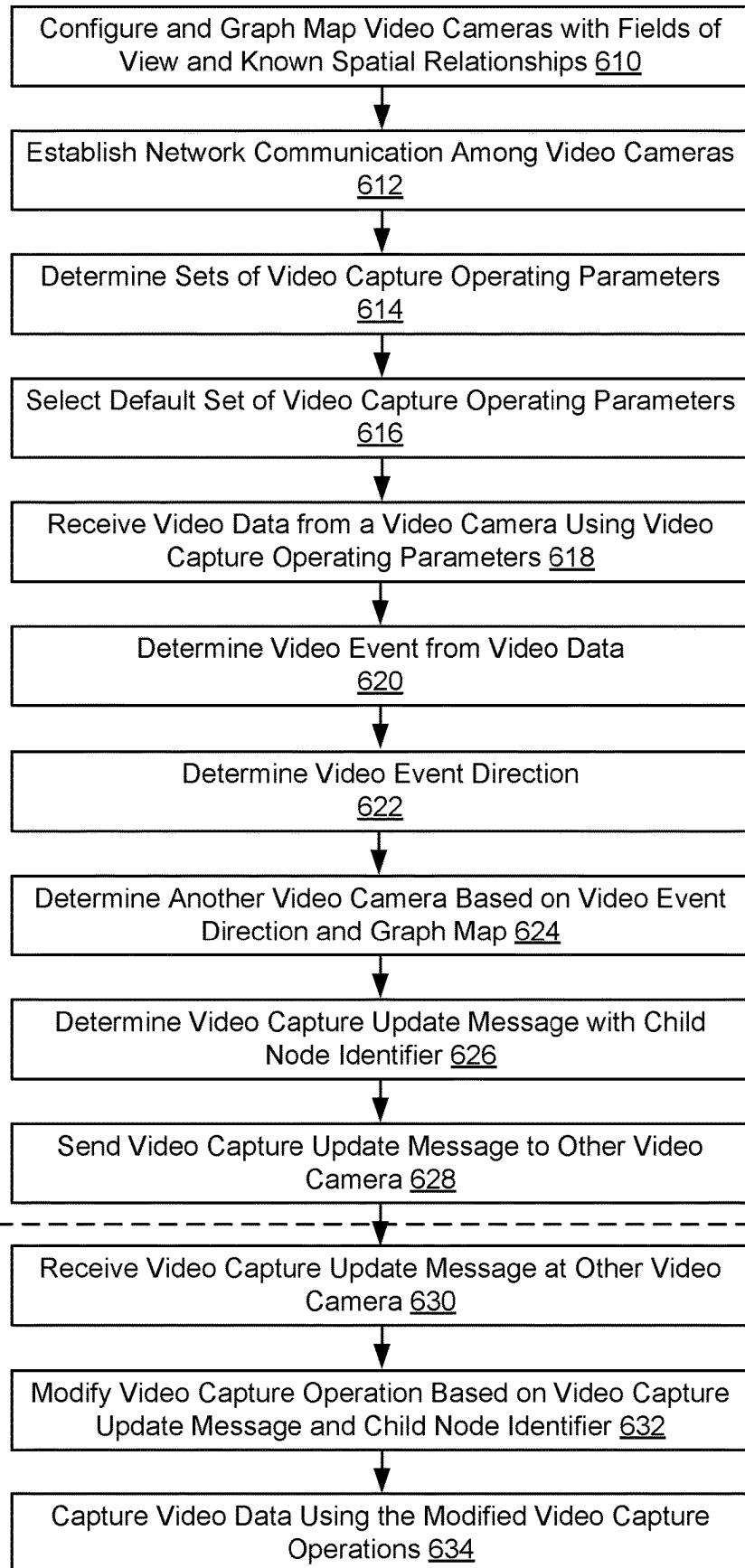
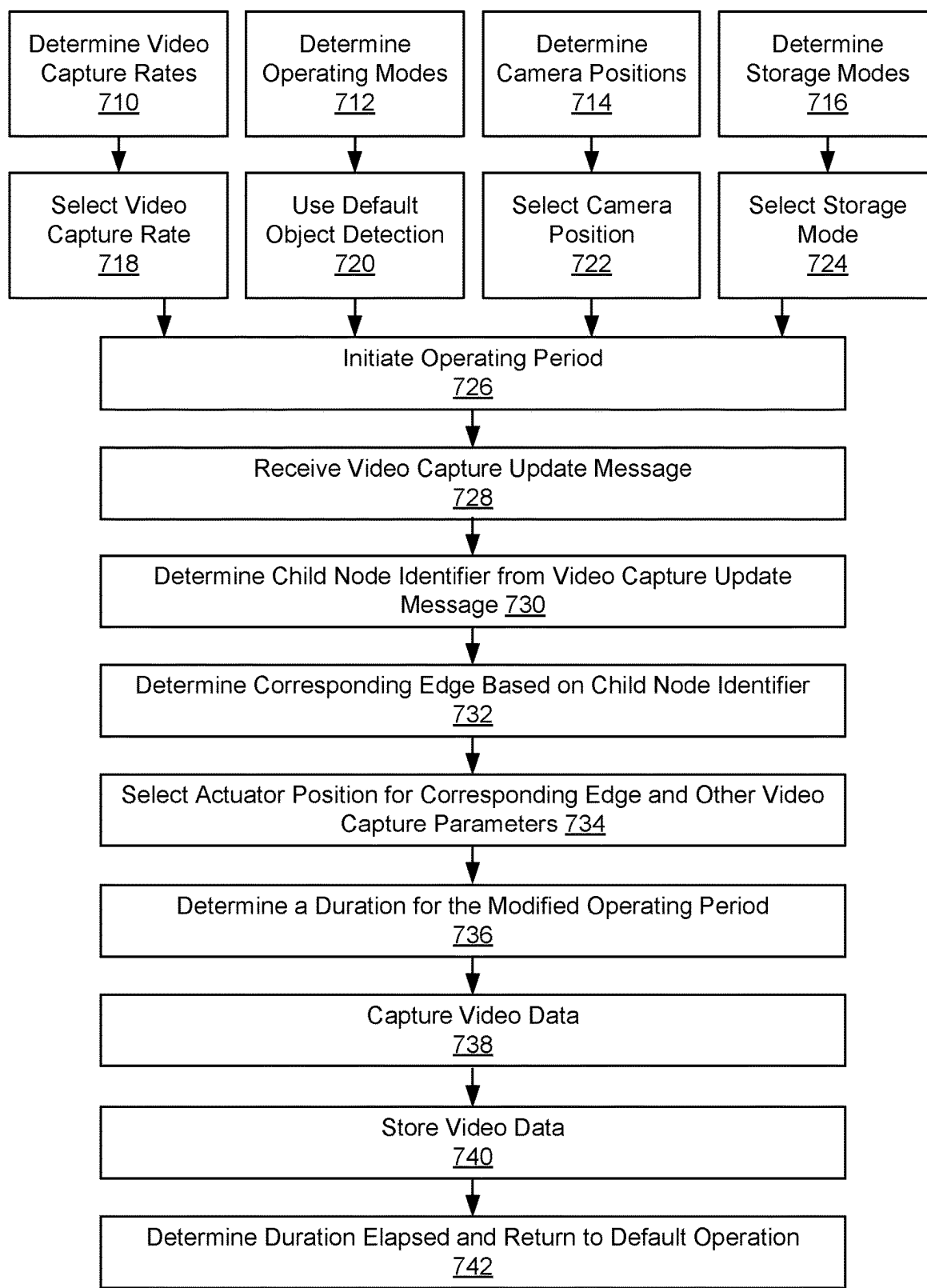


Figure 6



700

Figure 7

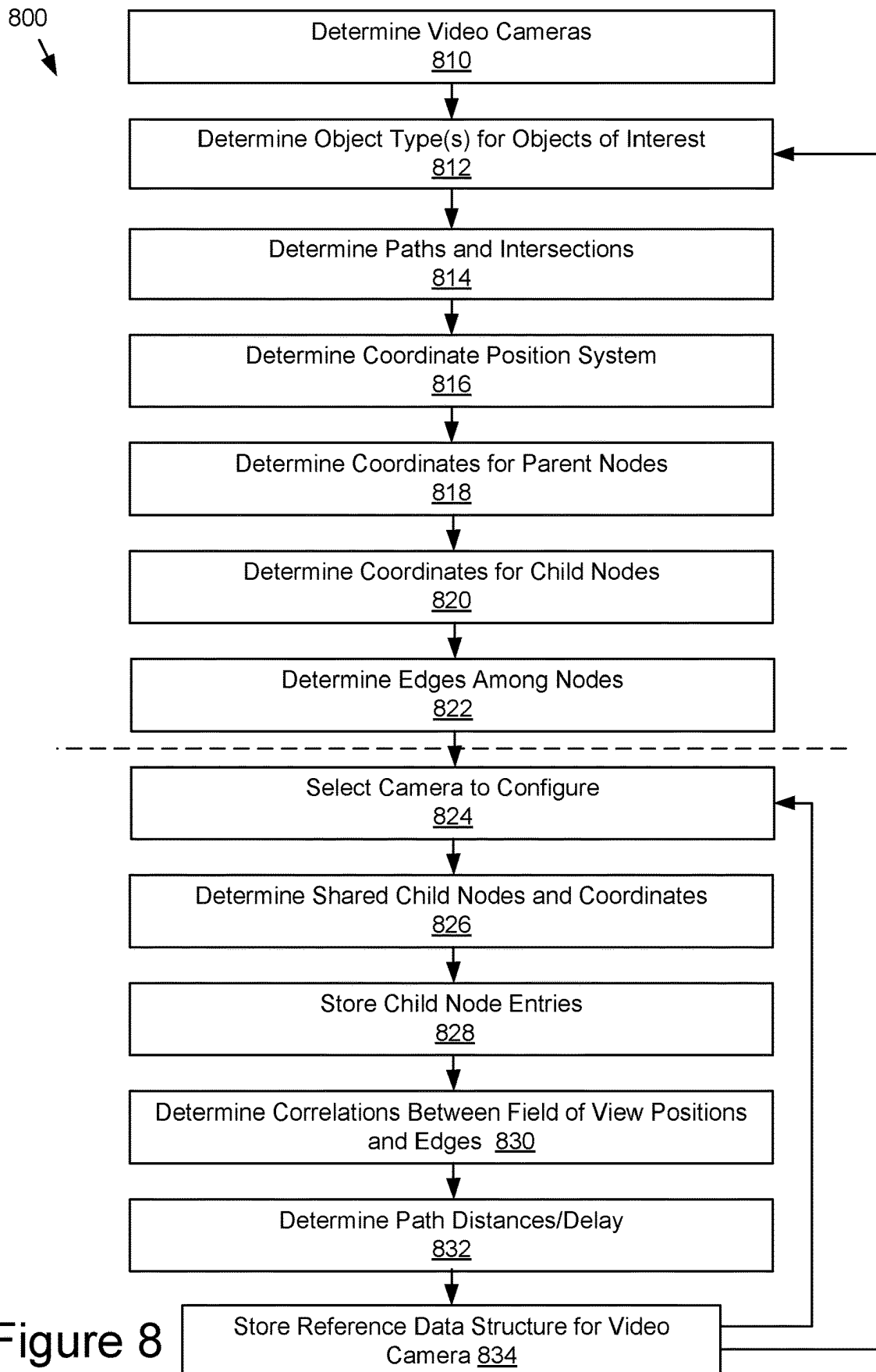


Figure 8

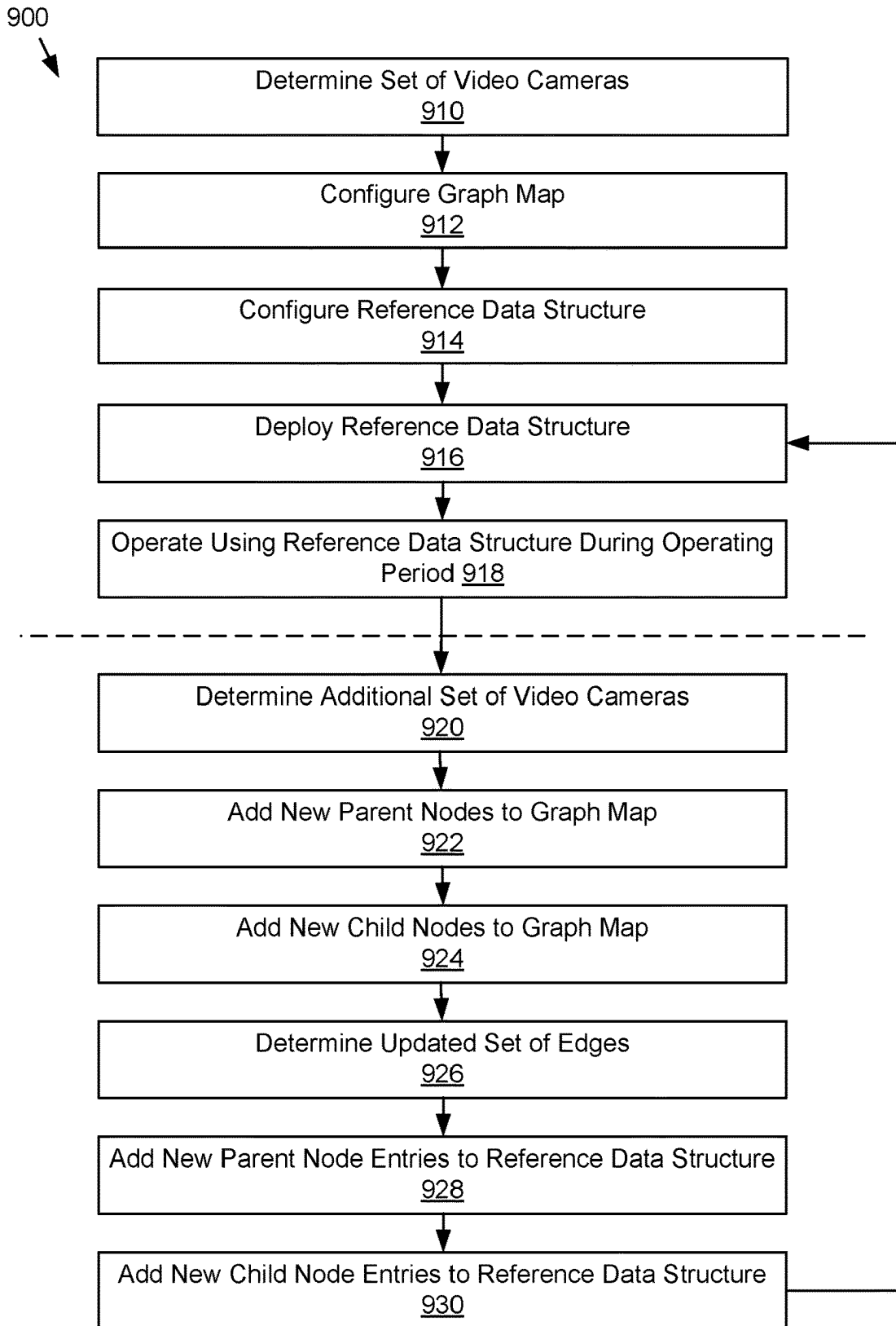


Figure 9

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PREDICTIVE ADJUSTMENT OF MULTI-CAMERA SURVEILLANCE VIDEO DATA CAPTURE USING GRAPH MAPS

TECHNICAL FIELD

The present disclosure generally relates to video surveillance systems and, more particularly, to video surveillance systems configured to control video capture parameters based on other networked video cameras.

BACKGROUND

Network-based video surveillance systems are a growing computing application in both business and personal markets. Some video surveillance systems may include one or more video cameras communicatively connected to a server, such as a network video recorder, through a wired interface, wired or wireless local area network, or wired or wireless wide area network, such as the internet. As video is recorded by the cameras, it is forwarded to the server system where it is stored and/or analyzed for subsequent retrieval. In some configurations, video may be recorded in onboard memory of the cameras, with or without transfer to a server. Client or user systems are communicatively connected to the server system and/or cameras to request, receive, and display streams of recorded video data and/or related alerts and analytics.

An increasing number of video surveillance applications use object detection and recognition, such as facial recognition, based on discrete objects identified in a video stream. Due to the high cost of storage for surveillance applications, including continuous surveillance, video cameras may include features to selectively capture high quality video. For example, once an object, motion, or similar visual condition is detected, the camera may modify its video capture operations to capture high quality data. These modifications of video capture operations may include initiating data capture, such as where a light or motion sensor provides the visual condition, changing the video capture rate from a lower video capture rate to a higher video capture rate, moving the field of view of the camera using zoom and/or pan-tilt-zoom (PTZ) capabilities, and/or changing data storage destinations.

While such trigger conditions reduce video data usage, they may also delay object detection, miss fast moving objects, miss data objects that are more difficult to detect at lower capture rates, miss objects within PTZ range that do not cross a current field of view, and/or miss early portions of a video event of interest (which may include critical angles, lighting, or other image features to support object recognition).

Systems and methods for predictively modifying video capture operations for selective capture of objects of interest based on data from another networked video camera in the neighborhood may be advantageous. A reliable and efficient way of using video data from a networked video camera in the same vicinity to initiate modification of video capture operations before an object is detectable in the field of view of the video camera may be needed.

SUMMARY

Various aspects for predictively modifying video capture operations in a multi-camera video surveillance network, particularly using video data from a first camera to initiate changes in video capture parameters of another camera with

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non-overlapping fields of view based on a graph map of the spatial relationships among the video cameras, are described.

One general aspect includes a system that includes a first video camera having a first field of view, where a plurality of networked video cameras includes the first video camera; and a controller in communication with the plurality of networked video cameras and configured to: receive video data from the first video camera; determine, from the video data, a first video event having an event direction; determine, based on the event direction and a graph map of the plurality of networked video cameras, at least one other video camera from the plurality of networked video cameras that shares a child node in the graph map with first video camera, where the child node is in the event direction; and send, to the at least one other video camera and based on the first video event, a video capture update message, where the video capture update message includes a child node identifier for the child node.

Implementations may include one or more of the following features. The system may include the plurality of networked video cameras, where each video camera of the at least one other video camera is configured to, responsive to receiving the video capture update message: determine the child node identifier from the video capture update message; modify, based on the child node identifier, a video capture operation of that other video camera; and capture, using that other video camera, video data based on the modified video capture operation. The at least one other video camera may include an actuator configured to adjust a direction of a field of view of the at least one other video camera, and modifying the video capture operation may include: operating, during a first operating period, in a first actuator position of a plurality of positions for adjusting the direction of the field of view; and selecting, responsive to the video capture update message and during a second operating period, a second actuator position of the plurality of positions. The graph map may include: a coordinate position system; a first set of coordinates for the at least one other video camera; and a second set of coordinates for the child node. The at least one other video camera may be further configured to determine, based on the first set of coordinates and the second set of coordinates, the second actuator position. Each video camera of the plurality of networked video cameras may be configured for a plurality of video capture rates, and modifying the video capture operation may include: operating, during a first operating period, at a first video capture rate from the plurality of video capture rates; and selecting, responsive to the video capture update message and during a second operating period, a second video capture rate to modify the video capture operation. The graph map may include a coordinate position system, a first set of coordinates for the at least one other video camera, a second set of coordinates for the child node, and a third set of coordinates for the first video camera; the at least one other video camera may include a plurality of sets of video capture operating parameters; and the at least one other video camera may be further configured to use a first set of video capture operating parameters during a first operating period. Modifying the video capture operation may include: determining, based on the graph map, a path distance from the first video camera to the at least one other video camera through the child node; determining, based on the path distance, a duration for a second operating period for determining, from video data of the at least one other video camera, a second video event related to the first video event; and selecting, responsive to the video capture update message and during the second

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operating period, a second set of video capture operating parameters. The first video camera may include a plurality of graph maps; determining the first video event may include detecting, using an object detection model, an object of interest in the video data; determining the at least one other video camera from the plurality of networked video cameras may include determining a set of object parameters for the object of interest and using the set of object parameters for the object of interest to select the graph map from the plurality of graph maps; and modifying the video capture operation of the at least one other video camera may occur prior to the object of interest entering a field of view of the at least one other video camera and may increase a likelihood of detecting the object of interest. The system may include a data structure, based on the graph map, including: a plurality of parent node entries for the plurality of networked video cameras; a plurality of child node entries for a plurality of child nodes corresponding to path intersections between at least two video cameras of the plurality of video cameras; and a set of coordinates, based on a coordinate position system of the graph map, for each child node entry of the plurality of child node entries, where the first video camera is further configured to use the data structure to determine, for the first video event, each shared child node in the event direction for the at least one other video camera. The data structure may further include a field of view position for each edge connecting a parent node to at least one child node and the first video camera may be further configured to: determine, based on the first field of view, the event direction and a corresponding field of view position; and determine, based on the corresponding field of view position, the child node. During a first operating period, the plurality of networked video cameras may include a first set of video cameras, the graph map may include a first set of child nodes among the first set of video cameras; and the data structure may include a first configuration of parent node entries, child node entries, and sets of coordinates. During a second operating period, the plurality of networked video cameras may include the first set of video cameras and a second set of video cameras, the graph map may include a second set of child nodes among the first set of video cameras and the second set of video cameras, and the data structure may include a second configuration that includes additional parent node entries for the second set of video cameras and additional child node entries and sets of coordinates for the second set of child nodes.

Another general aspect includes a computer-implemented method that includes: receiving video data from a first video camera of a plurality of networked video cameras; determining, from the video data, a first video event having an event direction; determining, based on the event direction and a graph map of the plurality of networked video cameras, at least one other video camera from the plurality of networked video cameras that shares a child node in the graph map with first video camera, where the child node is in the event direction; and sending, to the at least one other video camera and based on the first video event, a video capture update message, where the video capture update message includes a child node identifier for the child node.

Implementations may include one or more of the following features. The computer-implemented method may include, responsive to receiving the video capture update message: determining the child node identifier from the video capture update message; modifying, based on the child node identifier, a video capture operation of the at least one other video camera; and capturing, using the at least one other video camera, video data based on the modified video

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capture operation. The at least one other video camera may include an actuator configured to adjust a direction of a field of view of the at least one other video camera; and the graph map may include a coordinate position system, a first set of coordinates for the at least one other video camera, and a second set of coordinates for the child node. Modifying the video capture operation may include: operating, during a first operating period, in a first actuator position of a plurality of positions for adjusting the direction of the field of view; determining, responsive to the video capture update message and based on the first set of coordinates and the second set of coordinates, a second actuator position; and selecting, during a second operating period, the second actuator position of the plurality of positions. Each video camera of the plurality of networked video cameras may be configured for a plurality of video capture rates and modifying the video capture operation may include: operating, during a first operating period, at a first video capture rate from the plurality of video capture rates; and selecting, responsive to the video capture update message and during a second operating period, a second video capture rate to modify the video capture operation. The graph map may include a coordinate position system, a first set of coordinates for the at least one other video camera, a second set of coordinates for the child node, and a third set of coordinates for the first video camera; and the at least one other video camera may include a plurality of sets of video capture operating parameters. Modifying the video capture operation may include: using a first set of video capture operating parameters during a first operating period; determining, based on the graph map, a path distance from the first video camera to the at least one other video camera through the child node; determining, based on the path distance, a duration for a second operating period for determining, from video data of the at least one other video camera, a second video event related to the first video event; and selecting, responsive to the video capture update message and during the second operating period, a second set of video capture operating parameters. The first video camera may include a plurality of graph maps; determining the first video event may include detecting, using an object detection model, an object of interest in the video data; determining the at least one other video camera from the plurality of networked video cameras may include determining a set of object parameters for the object of interest, and using the set of object parameters for the object of interest to select the graph map from the plurality of graph maps; and modifying the video capture operation of the at least one other video camera may occur prior to the object of interest entering a field of view of the at least one other video camera and may increase a likelihood of detecting the object of interest. The computer-implemented method may include: configuring, based on the graph map, a data structure for the first video camera including a plurality of parent node entries for the plurality of networked video cameras, a plurality of child node entries for a plurality of child nodes corresponding to path intersections between at least two video cameras of the plurality of video cameras, and a set of coordinates, based on a coordinate position system of the graph map, for each child node entry of the plurality of child node entries; and determining, for the first video event and using the data structure, each shared child node in the event direction for the at least one other video camera. The computer-implemented method may include: determining, based on a field of view of the first video camera, the event direction and a corresponding field of view position, where the data structure may further include a field of view position for each edge connecting a

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parent node to at least one child node; and determining, based on the corresponding field of view position, the child node. The computer-implemented method may include configuring, during a first operating period, the data structure in a first configurations of parent node entries, child node entries, and sets of coordinates, where the plurality of networked video cameras may include a first set of video cameras during the first operating period and the graph map may include a first set of child nodes among the first set of video cameras in the first operating period. The computer-implemented method may include configuring, during a second operating period responsive to adding a second set of video cameras to the first set of video cameras in the plurality of networked video cameras, the data structure in a second configuration by: adding, to the graph map, a second set of parent nodes corresponding to the second set of video cameras; adding, to the graph map, a second set of child nodes among the first set of video cameras and the second set of video cameras; determining, for the graph map, an updated set of edges among the parent nodes and the child nodes; adding, to the data structure, additional parent node entries for the second set of video cameras; and adding, to the data structure, additional child node entries and sets of coordinates for the second set of child nodes.

Still another general aspect includes a system that includes: a plurality of networked video cameras; a processor; a memory; receiving video data from a first video camera of a plurality of networked video cameras; means for determining, from the video data, a first video event having an event direction; means for determining, based on the event direction and a graph map of the plurality of networked video cameras, at least one other video camera from the plurality of networked video cameras that shares a child node in the graph map with first video camera, where the child node is in the event direction; and means for sending, to the at least one other video camera and based on the first video event, a video capture update message, where the video capture update message includes a child node identifier for the child node.

The various embodiments advantageously apply the teachings of computer-based surveillance systems to improve the functionality of such computer systems. The various embodiments include operations to overcome or at least reduce the issues previously encountered in surveillance systems and, accordingly, are more effective and/or cost-efficient than other surveillance systems. That is, the various embodiments disclosed herein include hardware and/or software with functionality to improve selective capture of surveillance video data by using video data from other cameras in a vicinity to trigger modifications of video capture operations based on graph mapped relationships among the video cameras. Accordingly, the embodiments disclosed herein provide various improvements to network-based video surveillance systems.

It should be understood that language used in the present disclosure has been principally selected for readability and instructional purposes, and not to limit the scope of the subject matter disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 schematically illustrates a computer-based surveillance system.

FIG. 2 schematically illustrates an example graph map and camera configuration for multiple networked video cameras configured for using predictive adjustment to

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modify video capture operations, such as may be used by the computer-based surveillance system of FIG. 1.

FIG. 3 schematically illustrates some elements of the computer-based surveillance system of FIG. 1.

FIG. 4 is a flowchart of an example method of sending an alert based on graph mapped relationships among video cameras.

FIG. 5 is a flowchart of an example method of modifying video capture by repositioning based on the child node in an alert.

FIG. 6 is a flowchart of an example method of predictive modification of video capture operations among networked video cameras based on a video event detected by one camera modifying the operation of another camera.

FIG. 7 is a flowchart of an example method of modifying video capture operations based on a video capture update message from an event detected by another camera.

FIG. 8 is a flowchart of an example method of configuring multiple network video cameras for use of graph maps for managing spatial relationships and alerts.

FIG. 9 is a flowchart of an example method of configuring and updating graph maps and reference data structures for each video camera.

DETAILED DESCRIPTION

FIG. 1 shows an embodiment of an example video surveillance system **100** with multiple video cameras **110** interconnected to a network video recorder **130** for display of surveillance video on user device **170**. While some example features are illustrated, various other features have not been illustrated for the sake of brevity and so as not to obscure pertinent aspects of the example embodiments disclosed herein. In some embodiments, cameras **110**, network video recorder **130**, and user device **170** are computer-based components that may be interconnected by a network **102**. Additional components, such as network video server **160** and/or network video storage **162** may also be connected to network **102**. In some embodiments, one or more cameras may connect directly to network video recorder **130**, such as cameras **110.5-110.n** in FIG. 1, without communicating through network **102**. Similarly, in alternate embodiments (not shown), user device **170** may connect directly to network video recorder **130**.

In some embodiments, one or more networks **102** may be used to communicatively interconnect various components of surveillance system **100**. For example, each component, such as cameras **110**, network video recorder **130**, external storage device **140.n**, network video server **160**, network video storage **162**, and/or user device **170** may include one or more network interfaces and corresponding network protocols for communication over network **102**. Network **102** may include a wired and/or wireless network (e.g., public and/or private computer networks in any number and/or configuration) which may be coupled in a suitable way for transferring data. For example, network **102** may include any means of a conventional data communication network such as a local area network (LAN), a wide area network (WAN), a telephone network, such as the public switched telephone network (PSTN), an intranet, the internet, or any other suitable communication network or combination of communication networks. In some embodiments, network **102** may comprise a plurality of distinct networks, subnetworks, and/or virtual private networks (VPN) may be used to limit communications among specific components. For example, cameras **110** may be on a limited access network such that video and control data may only be

transmitted between cameras **110** and network video recorder **130**, enabling network video recorder **130** to control access to cameras **110** and their video data.

Cameras **110** may include analog or digital cameras connected to an encoder that generates an encoded video stream with a defined resolution, aspect ratio, and video encoding format. In some embodiments, cameras **110** may include internet protocol (IP) cameras configured to encode their respective video streams and stream them over network **102** to network video recorder **130**. In some embodiments (not shown), the encoder may reside in network video recorder **130**. In some embodiments, cameras **110** may be configured to receive audio data through integrated or connected microphones (not shown) and include embedded, synchronized, and/or independent audio streams with their respective video streams. In some embodiments, video cameras **110** may include an image sensor **112**, a processor or central processing unit CPU **114**, a memory **116**, an encoder **118**, an audio channel **120**, a control circuit **122**, and/or a network interface **126**. In some embodiments, video cameras **110** may include onboard analytics, such as a video analysis subsystem **124**.

For example, image sensor **112** may include a solid state device configured to capture light waves and/or other electromagnetic waves and convert the light into an image, generally composed of colored pixels. Image sensor **112** may determine a base image size, resolution, bandwidth, depth of field, dynamic range, and other parameters of the video image frames captured. Image sensor **112** may include charged couple device (CCD), complementary metal oxide semiconductor (CMOS), and/or other image sensor devices of various sensor sizes and aspect ratios. In some embodiments, image sensor **112** may be paired with one or more filters, such as infrared (IR) blocking filters, for modifying the light received by image sensor **112** and/or processed by camera **110**. For example, an IR blocking filter may be selectively enabled or disabled for different image capture use cases. In some embodiments, one or more video cameras **110** may include more than one image sensor and related video data paths. In some embodiments, multiple image sensors are supported by the same circuit board and/or processing subsystem containing CPU **114**, memory **116**, encoder **118**, audio channel **120**, control circuit **122**, analysis subsystem **124**, and/or network interface **126**.

Digital video data from image sensor **112** may be received by processor **114** for (temporary) storage and processing in memory **116** and/or encoding by encoder **118**. Processor **114** may include any type of conventional processor or micro-processor that interprets and executes instructions. In some embodiments, processor **114** may include a neural network processor, such as a neural network processor used by analysis subsystem **124** for supporting object recognition or other onboard analysis. Memory **116** may include a random access memory (RAM) or another type of dynamic storage device that stores information and instructions for execution by processor **114** and/or a read only memory (ROM) or another type of static storage device that stores static information and instructions for use by processor **114** and/or any suitable storage element such as a solid state storage element. Memory **116** may store basic input/output system (BIOS), firmware, and/or operating system instructions for initializing and executing the instructions and processes of cameras **110**. Encoder **118** may use various possible digital encoding and/or compression formats for encoding the video stream received by image sensor **112**. In some embodiments, encoder **118** may use a compressed video format to reduce the storage size and network bandwidth necessary for stor-

ing and transferring the original video stream. For example, encoder **118** may be configured to encode the video data as joint photographic expert group (JPEG), motion picture expert group (MPEG)-2, MPEG-4, advanced video coding (AVC)/H.264, and/or other video encoding standards or proprietary formats. In some configurations, the settings used by encoder **118** may be modified responsive to a video capture update message received by camera **110**.

Camera **110** may include audio channel **120** configured to capture audio data to be processed and encoded with image data in the resulting video stream and/or as an independent audio data stream. In some embodiments, one or more microphones may be enabled to capture audio data in parallel with the image data captured by image sensor **112** and/or may operate to capture audio data even when image sensor **112** is not actively capturing video data. For example, one or more microphones may be configured with an audio sensor that captures sound waves and converts them into a time-based audio data stream for an audio field in the vicinity of the video camera. In some embodiments, encoder **118** may include an audio encoder that operates in conjunction with the video encoder to encode a synchronized audio data stream in the video stream. For example, the video format used by encoder **118** may include one or more audio tracks for encoding audio data to accompany the image data during video stream playback. In some embodiments, encoder **118** may include a separate audio encoder for independently encoding an audio data stream that may be provided through audio channel **120** for data analysis independent of the video data.

Control circuit **122** may include a control circuit for managing the physical position of a camera **110**. In some embodiments, camera **110** may be a pan-tilt-zoom (PTZ) camera that is capable of remote directional and zoom control. Control circuit **122** may be configured to receive motion commands through network interface **126** and/or through another interface, such as a dedicated remote-control interface, such short distance infrared signals, Bluetooth, etc. For example, network video recorder **130** and/or user device **170** may be configured to send PTZ commands to control circuit **122**, which translates those commands into motor position control signals for a plurality of actuators that control the position of camera **110**. In some embodiments, control circuit **122** may include logic for automatically responding to movement or other triggers detected through image sensor **112** to redirect camera **110** toward the source of movement or other trigger. For example, an auto tracking feature may be embodied in firmware that enables the camera to estimate the size and position of an object based on changes in the pixels in the raw video stream from image sensor **112** and adjust the position of the camera to follow the moving object, returning to a default position when movement is no longer detected. Similarly, an auto capture feature may be embodied in firmware that enables the camera to determine and bound an object based on an object detection algorithm and center and zoom on that object to improve image size and quality. In some embodiments, control circuit **122** may include logic for virtual PTZ or ePTZ, which enables a high-resolution camera to digitally zoom and pan to portions of the image collected by image sensor **112**, with no physical movement of the camera. In some embodiments, control circuit **122** may include software and one or more application protocol interfaces (APIs) for enabling remote devices to control additional features and capabilities of camera **110**. For example, control circuit **122** may enable network video recorder **130**, another video camera **110**, and/or user device **170** to configure video

formats, enable and disable filters, set motion and/or audio detection, auto tracking, and similar features, and/or initiate video data streaming. In some embodiments, one or more systems may provide PTZ position control signals (and/or PTZ positioning commands converted to PTZ position control signals by control circuit 122) through the API. In some configurations, control circuit 122 may modify camera position or other features in response to a video capture update message received by camera 110.

In some embodiments, video camera 110 may include video analysis subsystem 124 configured for onboard video analytics. For example, video analysis subsystem 124 may be configured to use CPU 114 and memory 116 to execute at least a portion of video analytics for video data captured by video camera 110. In some embodiments, video analysis subsystem 124 may be configured to operate similarly to video analysis subsystem 156 in network video recorder 130, as further described below, and embody one or more analytics engines and/or analytical model libraries. In some embodiments, video analysis subsystem 124 may be configured to support object detection, classification, and/or recognition. For example, video analysis subsystem 124 may receive the real-time video data stream from sensor 112 and/or encoder 118, detect a video event, and instruct another camera to modify video capture operations prior to an object of interest entering the field of view of that other camera. In some configurations, the settings and reference data used by video analysis subsystem 124 may be modified responsive to a video capture update message received by camera 110.

Network interface 126 may include one or more wired or wireless connections to network 102 and/or a dedicated camera interface of network video recorder 130. For example, network interface 126 may include an ethernet jack and corresponding protocols for IP communication with network video recorder 130. In some embodiments, network interface 126 may include a power over ethernet (POE) connection with network video recorder 130 or another camera access point. PoE may enable both power for camera 110 and network data to travel on the same wire. In some embodiments, network interface 126 may enable an IP camera to be configured as a network resource with an IP address that is accessible on a LAN, WAN, or the internet. For example, network video recorder 130 and/or user device 170 may be configured to selectively receive video from cameras 110 from any internet-connected location using internet addressing and security protocols.

Network video recorder 130 may include a computer system configured as a video storage device to record the video streams from cameras 110. For example, network video recorder 130 may be configured to receive video streams from each of cameras 110 for storage, analysis, and/or display through user device 170. In some embodiments, cameras 110 may send encoded video streams based on the raw image data collected from their respective image sensors 112, with or without video data compression. A single video stream may be received from each camera 110 and network video recorder 130 may be configured to receive video streams from all connected cameras in parallel, as network bandwidth and processing resources allow.

Network video recorder 130 may include a housing and a bus interconnecting at least one processor or CPU 132, at least one memory 134, at least one storage device 140, and at least one interface, such as camera interface 136, network interface 138, and/or storage interface 142. The housing (not shown) may include an enclosure for mounting the various subcomponents of network video recorder 130, locating any

physical connectors for the interfaces, and protecting the subcomponents. Some housings may be configured for mounting within a rack system. The bus (not shown) may include one or more conductors that permit communication among the components of network video recorder 130. Processor 132 may include any type of processor or micro-processor that interprets and executes instructions or operations. Memory 134 may include a random access memory (RAM) or another type of dynamic storage device that stores information and instructions for execution by processor 132 and/or a read only memory (ROM) or another type of static storage device that stores static information and instructions for use by processor 132 and/or any suitable storage element.

In some embodiments, network video recorder 130 may include camera interface 136 configured for connection with one or more cameras 110. For example, camera interface 136 may include a plurality of ethernet ports and supporting protocols compatible with PoE standards for connecting to cameras 110.5-110.n. In some embodiments, camera interface 136 may include a PoE network switch for providing power to connected cameras and routing data packets to and from cameras 110.5-110.n, such as control and video data. In some embodiments, network video recorder 130 may not include a dedicated camera interface 136 and may use network interface 138 for communication with cameras 110 over network 102.

Network interface 138 may include one or more wired or wireless network connections to network 102. Network interface 138 may include a physical interface, such as an ethernet port, and related hardware and software protocols for communication over network 102, such as a network interface card.

Storage devices 140 may include one or more non-volatile memory devices configured to store video data, such as a hard disk drive (HDD) or solid state drive (SSD). In some embodiments, storage device 140 is, or includes, a plurality of solid-state drives. In some embodiments, network video recorder 130 may include internal storage device 140.1 and expandable storage that enables additional storage devices 140.n to be connected via storage interface 142. Each storage device 140 may include a non-volatile memory (NVM) or device controller 144 based on compute resources (processor and memory) and a plurality of NVM or media devices 146 for data storage (e.g., one or more NVM device(s), such as one or more flash memory devices). In some embodiments, a respective data storage device 140 of the one or more data storage devices includes one or more NVM controllers, such as flash controllers or channel controllers (e.g., for storage devices having NVM devices in multiple memory channels). In some embodiments, storage devices 140 may each be packaged in a housing, such as a multi-part sealed housing with a defined form factor and ports and/or connectors for interconnecting with storage interface 142. Storage device 140.1 and each expanded storage device 140.n may be of the same storage device type or a different storage device type.

In some embodiments, a respective data storage device 140 may include a single non-volatile storage medium device, while in other embodiments the respective data storage device 140 includes a plurality of media devices. In some embodiments, media devices include NAND-type flash memory or NOR-type flash memory. In some embodiments, storage device 140 may include one or more hard disk drives. In some embodiments, storage devices 140 may include a flash memory device, which in turn includes one or more flash memory die, one or more flash memory

packages, one or more flash memory channels or the like. However, in some embodiments, one or more of the data storage devices **140** may have other types of non-volatile data storage media (e.g., phase-change random access memory (PCRAM), resistive random access memory (ReRAM), spin-transfer torque random access memory (STT-RAM), magneto-resistive random access memory (MRAM), etc.).

In some embodiments, each storage device **140** includes a device controller **144**, which includes one or more processing units (also sometimes called CPUs or processors or microprocessors or microcontrollers) configured to execute instructions in one or more programs. In some embodiments, the one or more processors are shared by one or more components within, and in some cases, beyond the function of the device controllers. Media devices **146** are coupled to device controllers **144** through connections that typically convey commands in addition to data, and optionally convey metadata, error correction information and/or other information in addition to data values to be stored in media devices and data values read from media devices **146**. Media devices **146** may include any number (i.e., one or more) of memory devices including, without limitation, non-volatile semiconductor memory devices, such as flash memory device(s). In some embodiments, media devices **146** may include NAND or NOR flash memory devices comprised of single level cells (SLC), multiple level cell (MLC), triple-level cells, or more.

In some embodiments, media devices **146** in storage devices **140** are divided into a number of addressable and individually selectable blocks, sometimes called erase blocks. In some embodiments, individually selectable blocks are the minimum size erasable units in a flash memory device. In other words, each block contains the minimum number of memory cells that can be erased simultaneously (i.e., in a single erase operation). Each block is usually further divided into a plurality of pages and/or word lines, where each page or word line is typically an instance of the smallest individually accessible (readable) portion in a block. In some embodiments (e.g., using some types of flash memory), the smallest individually accessible unit of a data set, however, is a sector or codeword, which is a subunit of a page. That is, a block includes a plurality of pages, each page contains a plurality of sectors or codewords, and each sector or codeword is the minimum unit of data for reading data from the flash memory device.

A data unit may describe any size allocation of data, such as host block, data object, sector, page, multi-plane page, erase/programming block, media device/package, etc. Storage locations may include physical and/or logical locations on storage devices **140** and may be described and/or allocated at different levels of granularity depending on the storage medium, storage device/system configuration, and/or context. For example, storage locations may be allocated at a host logical block address (LBA) data unit size and addressability for host read/write purposes but managed as pages with storage device addressing managed in the media flash translation layer (FTL) in other contexts. Media segments may include physical storage locations on storage devices **140**, which may also correspond to one or more logical storage locations. In some embodiments, media segments may include a continuous series of physical storage location, such as adjacent data units on a storage medium, and, for flash memory devices, may correspond to one or more media erase or programming blocks. A logical data group may include a plurality of logical data units that may be grouped on a logical basis, regardless of storage

location, such as data objects, video media files, or other logical data constructs composed of multiple host blocks. In some embodiments, storage device **140** may be configured specifically for managing the storage and overwriting of video data in a continual monitoring application for video surveillance.

Storage interface **142** may include a physical interface for connecting to one or more external storage devices using an interface protocol that supports storage device access. For example, storage interface **142** may include a peripheral component interconnect express (PCIe), serial advanced technology attachment (SATA), small computer system interface (SCSI), serial attached SCSI (SAS), universal serial bus (USB), Firewire, or similar storage interface connector supporting storage protocol access to storage devices **140.n**. In some embodiments, storage interface **142** may include a wireless data connection with sufficient bandwidth for video data transfer. Depending on the configuration and protocols used by storage interface **142**, storage device **140.n** may include a corresponding interface adapter, firmware, and/or protocols for receiving, managing, and responding to storage commands from network video recorder **130**.

Network video recorder **130** may include a plurality of modules or subsystems that are stored and/or instantiated in memory **134** for execution by processor **132** as instructions or operations. For example, memory **134** may include a camera control subsystem **150** configured to control cameras **110**. Memory **134** may include a video capture subsystem **152** configured to receive video streams from cameras **110**. Memory **134** may include a video storage subsystem **154** configured to store received video data in storage device(s) **140** and/or network video storage **162**. Memory **134** may include a video analysis subsystem configured to analyze video streams and/or video data for defined events, such as motion, recognized objects, recognized faces, and combinations thereof. Memory **134** may include a video display subsystem configured to selectively display video streams on user device **170**, which may be attached to network video recorder **130** or remotely connected via network **102**.

In some embodiments, camera control subsystem **150** may include interface protocols and a set of functions and parameters for using, configuring, communicating with, and providing command messages to cameras **110**. For example, camera control subsystem **150** may include an API and command set for interacting with control circuit **122** to access one or more camera functions. In some embodiments, camera control subsystem **150** may be configured to set video configuration parameters for image sensor **112** and/or video encoder **118**, access pan-tilt-zoom features of control circuit **122**, set or modify camera-based motion detection, tripwire, and/or low light detection parameters in memory **116**, and/or otherwise manage operation of cameras **110**. For example, camera control subsystem **150** may maintain a video camera configuration table, pages, or similar data structures that includes entries for each video camera being managed and their respective camera-specific configuration parameters, active control features (such as PTZ control), and other configuration and control information for managing cameras **110**. In some embodiments, each camera **110** may be assigned a unique camera identifier that may be used by camera control subsystem **150**, video capture subsystem **152**, and/or other subsystems to associate video data with the camera from which it was received. In some configurations, network video recorder **130** may be configured to respond to video events detected from the video stream for one camera by sending camera control messages, such as a video capture

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update message, to modify the video capture of other cameras through camera control subsystem **150**. For example, camera control subsystem **150** may include multiple camera-specific sets of configuration parameters for video capture operations and a neighborhood table for determining how each other camera should respond to video events detected from another camera.

In some embodiments, video capture subsystem **152** may include interface protocols and a set of functions and parameters for receiving video streams from cameras **110**. For example, video capture subsystem **152** may include video data channels and related data buffers for managing a plurality of camera video data streams. In some embodiments, each video camera **110** may be allocated a dedicated video channel for continuously and/or selectively sending its video stream to network video recorder **130**. Video capture subsystem **152** may be configured to pass each received video stream to video storage subsystem **154**, video analysis subsystem **156**, and/or video display subsystem **158**. For example, received video streams may be buffered by video capture subsystem **152** before being streamed to video storage subsystem **154** and split into dual video streams with different video parameters for video analysis subsystem **156** and video display subsystem **158**.

In some embodiments, video storage subsystem **154** may include interface protocols and a set of functions and parameters for managing storage of video data in storage devices **140** and/or network video storage **162** for later retrieval and use by video analysis subsystem **156** and/or video display subsystem **158**. For example, video storage subsystem **154** may write camera video stream data from video data buffers to non-volatile storage in storage devices **140** and video analysis subsystem **156** and/or video display subsystem **158** may be configured to selectively read video data from storage devices **140**. In some embodiments, video storage subsystem **154** may include management of video storage space in storage devices **140** and/or network video storage **162** in accordance with one or more data retention and/or data archiving schemes. For example, surveillance system **100** may support continuous and/or triggered recording of video data from cameras **110** and video storage subsystem **154** may include logic for enforcing a data retention and overwriting policy whereby the fixed storage space of storage devices **140** is recycled for storing a recent period of captured video, video data meeting specific retention criteria, and/or deleting or archiving video data after one or more periods of time defined in the data retention policy. In some embodiments, video storage subsystem **154** may include or access video decoders and/or encoders for storing video data in a storage video format that is different than the camera video format, such as using a different codec, compression factor, frame rate, resolution, image size, etc. In some embodiments, video storage subsystem may store embedded audio data with the video data and/or may store audio data in separate audio data files or objects.

In some embodiments, video analysis subsystem **156** may include interface protocols and a set of functions and parameters for analyzing video data from cameras **110**. For example, video analysis subsystem **156** may be configured to run one or more event detection algorithms for determining, tagging, and/or initiating alerts or other actions in response to detected video events and/or audio events. In some embodiments, video analysis subsystem **156** may be configured to tag or build metadata structures that map detected events to time and image location markers for the video stream from which they are detected. For example, video analysis subsystem **156** may use motion, tripwire,

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object recognition, facial recognition, audio detection, speech recognition, and/or other algorithms to determine events occurring in a video stream and tag them in a corresponding metadata track and/or separate metadata table associated with the video data object. In some embodiments, video analysis subsystem **156** may include event handling logic for determining response to detection of one or more detected events, such as raising an alert to user device **170** or triggering selective display of a video stream including the detected event through video display subsystem **158**. In some embodiments, video analysis subsystem **156** may operate in real-time or near real-time on video data received by video capture subsystem **152**, delayed processing of video data stored by video storage subsystem **154**, and/or a combination thereof based on the nature (and processing requirements) of the video events, volume of video to be processed, and other factors. In some embodiments, video analysis subsystem **156** may comprise one or more analytics engines configured for a particular type of event and corresponding event detection algorithm or model.

In some embodiments, video display subsystem **158** may include interface protocols and a set of functions and parameters for displaying video from video capture subsystem **152** and/or video storage subsystem **154** on user device **170**. For example, video display subsystem **158** may include a monitoring or display configuration for displaying one or more video streams in real-time or near real-time on a graphical user display of user device **170** and/or receive video navigation commands from user device **170** to selectively display stored video data from video storage subsystem **154**. In some embodiments, video display subsystem **158** may maintain an index of real-time/near real-time video streams and/or stored or archived video streams that are available for access by user device **170**. In some embodiments, the video index may include a corresponding metadata index that includes video data parameters (e.g., time, location, camera identifier, format, low light/normal light, etc.), detected video event metadata (event time, location, type, parameters, etc.), and/or video management parameters (expiration, active/archive, access control, etc.) for use in displaying and managing video data. Video display subsystem **158** may be configured to support user device **170** when directly attached to network video recorder **130** and/or via network **102** within a LAN, WAN, VPN, or the internet.

In some embodiments, surveillance system **100** may include one or more remote and/or cloud-based resources for supporting the functions of network video recorder **130** and/or user device **170**. For example, surveillance system **100** may include a network video server **160** configured to host some, all, or select portions of the functions of network video recorder **130**, such as a cloud-based server system. As another example, surveillance system **100** may include network video storage **162** for storing active and/or archived video data, supplementing and/or replacing storage devices **140**, such as a cloud-based network attached storage system or distributed storage system. In some embodiments, the majority of functions described above for network video recorder **130** may reside in network video recorder **130** and select functions may be configured to leverage additional resources in network video server **160** and/or network video storage **162**. For example, network video server **160** may be configured to support specialized and/or processing intensive event detection algorithms to supplement video analysis subsystem **156** and/or network video storage **162** may be configured to support archiving of inactive video data for longer term storage.

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User device **170** may be any suitable computer device, such as a computer, a computer server, a laptop computer, a tablet device, a netbook, an internet kiosk, a personal digital assistant, a mobile phone, a smart phone, a gaming device, or any other computing device. User device **170** is sometimes called a host, client, or client system. In some embodiments, user device **170** may host or instantiate one or more applications for interfacing with surveillance system **100**. For example, user device **170** may be a personal computer or mobile device running a surveillance monitoring and management application configured to provide a user interface for network video recorder **130**. In some embodiments, user device **170** may be configured to access cameras **110** and/or their respective video streams through network video recorder **130** and/or directly through network **102**. In some

embodiments, one or more functions of network video recorder **130** may be instantiated in user device **170** and/or one or more functions of user device **170** may be instantiated in network video recorder **130**. User device **170** may include one or more processors or CPUs **172** for executing compute operations or instructions stored in memory **174** for accessing video data and other functions of network video recorder **130** through network **102**. In some embodiments, processor **172** may be associated with memory **174** and input/output device **176** for executing both video display operations and surveillance system management operations. Processor **172** may include any type of processor or microprocessor that interprets and executes instructions or operations. Memory **174** may include a random access memory (RAM) or another type of dynamic storage device that stores information and instructions for execution by processor **172** and/or a read only memory (ROM) or another type of static storage device that stores static information and instructions for use by processor **172** and/or any suitable storage element. In some embodiments, user device **170** may allocate a portion of memory **174** and/or another local storage device (in or attached to user device **170**) for storing selected video data for user device **170**. In some embodiments, user device **170** may include one or more input/output (I/O) devices **176**. For example, a graphical display, such as a monitor and/or touch screen display, and/or other user interface components such as a keyboard, a mouse, function buttons, speakers, vibration motor, a track-pad, a pen, voice recognition, biometric mechanisms, and/or any number of supplemental devices to add functionality to user device **170**. Network interface **178** may include one or more wired or wireless network connections to network **102**. Network interface **178** may include a physical interface, such as an ethernet port, and/or related hardware and software protocols for communication over network **102**, such as a network interface card, wireless network adapter, and/or cellular data interface.

User device **170** may include a plurality of modules or subsystems that are stored and/or instantiated in memory **174** for execution by processor **172** as instructions or operations. For example, memory **174** may include a video manager **180** configured to provide a user interface for selectively navigating and displaying real-time, near real-time, and/or stored video streams. Memory **174** may include alert manager **182** configured to provide a user interface for setting, monitoring, and displaying alerts based on video events. Memory **174** may include a camera manager **184** configured to provide a user interface for identifying, configuring, and managing cameras **110**. Memory **174** may include a configuration manager **186** to provide a user interface for setting and managing system settings, user access controls, storage options, and other configuration

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settings for surveillance system **100**. Memory **174** may include a network video recorder manager **188** configured to provide a user interface for identifying, configuring, and managing network video recorder **130** and/or multiple network video recorders. Memory **174** may include an analytics manager configured to provide a user interface for selecting, training, and managing event detection algorithms for surveillance system **100**.

In some embodiments, video manager **180** may include interface protocols and a set of functions and parameters for navigating and displaying video streams from cameras **110**. For example, video manager **180** may include a graphical user interface and interactive controls for displaying lists, tables, thumbnails, or similar interface elements for selecting and displaying video streams for particular cameras, times, locations, and/or events. In some embodiments, video manager **180** may enable split screen display of multiple camera video streams. For example, the near real-time video streams (with a predetermined lag based on network lag, storage, and processing times) from all active cameras may be displayed on a monitoring interface or a set of video streams corresponding to a detected event may be displayed in an event review interface. In some embodiments, video manager **180** may include a data structure summarizing all video data stored in surveillance system **100** to enable the user to locate and view older surveillance video data. For example, a video management log or database may include entries for stored video data indexed by related metadata, such as video data parameters (e.g., time, location, camera identifier, format, low light/normal light, etc.), detected video event metadata (event time, location, type, parameters, etc.), and/or video management parameters (expiration, active/archive, access control, etc.). In some embodiments, video manager **180** may be configured to interface with video display subsystem **158** and/or storage subsystem **154** for determining and retrieving selected video data.

In some embodiments, alert manager **182** may include interface protocols and a set of functions and parameters for setting, monitoring, and displaying alerts based on video events. For example, the user may define a set of trigger events that generate visual, audible, tactile, and/or notification-based (electronic mail, text message, automated call, etc.) alert to user device **170**. In some embodiments, alert manager **182** may include a plurality of preset alert conditions with associated event parameters and allow a user to enable and disable alert types and/or change associated event parameters. In some embodiments, alert manager **182** may be configured to overlay graphical elements representing detected events or event indicators on video streams displayed through video manager **180**. For example, detected motion, objects, or faces may be boxed or highlighted, tagged with relevant identifiers, or otherwise indicated in the video playback on user device **170**. In some embodiments, alert manager **182** may be configured to interface with video analysis subsystem **156**, video capture subsystem **152**, and/or directly with cameras **110** for receiving event notifications or parameters.

In some embodiments, camera manager **184** may include interface protocols and a set of functions and parameters for identifying, configuring, and managing cameras **110**. Configuration manager **186** may include interface protocols and a set of functions and parameters for setting and managing system settings, user access controls, storage options, and other configuration settings. Network video recorder (NVR) manager **188** may include interface protocols and a set of functions and parameters for identifying, configuring, and managing network video recorder **130**. For example, each of

camera manager **184**, configuration manager **186**, and/or NVR manager **188** may include a series of graphical user interfaces for displaying their respective component identifiers and related configuration parameters and enabling the user to view and/or change those parameters for managing surveillance system **100** and its component systems. In some embodiments, camera manager **184**, configuration manager **186**, and/or NVR manager **188** may provide changes parameters to the effected components, such as camera manager **184** sending camera configuration parameter changes to selected cameras **110**, NVR manager **188** sending NVR configuration parameter changes to network video recorder **130**, and/or configuration manager **186** sending system configuration parameter changes to all effected components.

In some embodiments, analytics manager **190** may include interface protocols and a set of functions and parameters for selecting, training, and managing event detection algorithms. For example, analytics manager **190** may include a library of event detection algorithms for different event types. In some embodiments, the event detection algorithms may include a set of parameters and/or model weights that are preconfigured based on training data sets processed independent of surveillance system **100**. For example, analytics manager **190** may include object detection algorithms for common objects, situations, and camera configurations. In some embodiments, analytics manager **190** may include preconfigured training data sets and/or allow the user to define training data sets for determining or refining event detection algorithm parameters and/or model weights based on predefined base algorithms or models. In some embodiments, analytics manager **190** may interface with analysis subsystem **156** for using the event detection algorithms configured through analytics manager **190** to process video data received by network video recorder **130** and/or selecting, training, and managing those algorithms.

FIG. **2** shows schematic representations of multiple networked video cameras configured for using predictive adjustment to modify video capture operations, implemented in a computer-based surveillance system **200**, such as surveillance system **100** in FIG. **1**. In the example configurations shown, cameras **110** are configured for onboard capture and analysis of video data, but similar functions may be implemented through a host, such as network video recorder **130** described above. In camera environment **202**, the three networked video cameras **110.1-110.3** are located around plurality of locations with interconnected paths, such as a network of roads and/or walking paths in city, campus, park, or similar location. In one example configuration, each camera **110** may be placed on the exterior of a building or mounted to a municipal service pole or similar structure to provide video capture in the vicinity of the building, intersection, or other point of interest. Upon installation, cameras **110** have known locations with known spatial relationships to one another and defined fields of view **206**. For example, a generally overhead view of camera environment **202** may allow cameras **110** and other relevant attributes of camera environment **202** to be graph mapped using a coordinate position system. Coordinate position systems may include two-dimensional mapping based on longitude and latitude or another X-Y coordinate system and scale applied to camera environment **202**. In some configurations, a three-dimensional mapping may be used, such as adding elevation, altitude, or a similar Z coordinate to the two-dimensional coordinate position system. Fields of view **206** may be defined in terms of two or more bounding vectors and may include default position

and maximum ranges in each direction of adjustment if equipped with PTZ actuators.

As shown in FIG. **2**, a graph map (using a two-dimensional coordinate position system in the example shown) may be overlaid on camera environment **202**. Each camera **110** may be treated as a parent node **204**. For example, the coordinate position of video camera **110.1** may be abstracted as parent node **204.1**, the coordinate position of video camera **110.2** may be abstracted as parent node **204.2**, and the coordinate position of video camera **110.3** may be abstracted as parent node **204.3**. Paths of travel between or among cameras **110** may be mapped and intersections may be abstracted as child nodes **208** (or sub-nodes). Each child node **208** may be a child node of each parent node that it connects to without going through another parent node or requiring a turn at an intersection (child nodes are connected to parent nodes by straight paths that may pass through another child node, such as node **208.3** being a child node of parent node **204.1** but not parent node **204.2**). For example, child nodes **208.1**, **208.2**, **208.3**, and **208.5** may be child nodes of camera **110.1**, child nodes **208.1**, **208.2**, **208.4**, and **208.5** may be child nodes of camera **110.2**, and child nodes **208.1**, **208.3**, and **208.4** may be child nodes of camera **110.3**. Each path segment between nodes represents an edge in the graph map. For example, edge **210.1** extends from parent node **204.1** to child node **208.1**, edge **210.2** extends from child node **208.1** to child node **208.2**, and so on for edges **210.3**, **210.4**, **210.5**, **210.6**, **210.7**, **210.8**, **210.9**, and **210.10**. Possible object paths may be comprised of each set of edges from one parent node to another parent node, generally through one or more child nodes. In some configurations, the presence of one or more child nodes between each pair of parent nodes may indicate that the cameras do not have overlapping fields of view and may not be capable of simultaneously capturing the same object of interest.

Cameras **110** may be deployed to monitor their respective fields of view **206** for object detection and related object detection event handling, such as alerts, selective archival storage, etc. For example, cameras **110** may be placed in their respective surveillance locations to detect object of interest **240** (in this case, a person) when it moves into their respective fields of view **206** along path of movement **242**. Other example objects of interest may include vehicles, animals, equipment, etc. Path of movement **242** may indicate an event direction from the camera that has most recently captured the object of interest (e.g., camera **110.1**). However, the presence of intersections and decision points modeled by child nodes **208** may reflect that object of interest **240** may not travel in a straight path toward another camera, and yet may ultimately enter field of view **206** for other cameras based on available object paths. For example, object of interest **240** may enter field of view **206.2** of camera **110.2** by traveling to child node **208.2** and turning right to follow edge **210.3**. Alternatively, object of interest **240** may continue straight at child node **208.2** and turn right at child node **208.3** to follow edge **210.5** toward camera **110.3**. However, unless camera **110.3** proactively changes its actuator position and field of view **206.3**, it is unlikely to capture object of interest **240**.

In some configurations, cameras **110** may be configured for video capture **212** based on different video capture rates. For example, cameras **110** may include a passive video capture rate **212.1** that includes lower quality video (e.g., lower pixel count and/or frame rate) to save storage space and/or network bandwidth for operating periods when no objects of interest are likely to be present in the field of view. Cameras **110** may include an active video capture rate **212.2**

that includes higher quality video (e.g., higher pixel count and/or frame rate) to provide better video data for analysis, display, and/or storage of possible object detection events. So, even if camera **110.2** is already oriented toward the most likely object travel path, it may be advantageous to proactively and predictively trigger a change in video capture **212**.

Once camera **110.1** has detected object of interest **240**, it may be advantageous to predictively and selectively adjust the video capture operations of the other cameras. While it may be possible to alert all cameras to modify their video capture operations, doing so every time any of the cameras detected an object of interest would result in a waste of storage, network bandwidth, and other resources. In some configurations, competition for constrained resources in the system, such as limited network bandwidth and/or storage channels, may prevent all cameras from operating at active capture rate **212.2** at the same time and/or for extended periods. Therefore, prediction of which cameras are most likely to have the object enter its field of view may be desirable and to allow those cameras to modify their video capture operations to maximize the likelihood of capturing the object without wasting resources. If camera **110.3** modifies its video capture operations, such as by switching to its active capture rate, pivoting to its right facing actuator position, prioritizing the object detector for the object type of the object of interest, and other changes of video capture operating parameters, before object of interest **240** enters field of view **206.3**, it may maximize the chances of capturing high-quality images of the object to support further detection, recognition, monitoring, or other analysis.

Cameras **110** may have been deployed with object detectors **214** configured to detect objects of interest using video data captured at their passive capture rate **212.1** and/or active capture rate **212.2**. For example, object detectors **214** may be trained on low-quality video to detect one or more objects of interest at a relatively low confidence threshold to trigger active capture rate **212.2** and processing of additional video data at the higher quality and with an object detector trained on higher quality data to confirm object detection, enable classification and/or object recognition, or support other analysis. Alternately, other sensors, such as motion sensors, audio sensors, etc. or detection algorithms, such as video tripwires, etc., may be used to initially trigger active capture rate **212.2** for a camera when an object of interest initially enters camera environment **202**. In some configurations, additional processing of the high-quality video data may include classification of the object type and/or determination of a direction of travel for the object of interest.

Cameras **110** may include trigger logic **216** for determining a video event that triggers predictive adjustment of other video cameras. For example, trigger logic **216** may be based on one or more parameters of the detected object and/or subsequent processing. In some configurations, objects of specific types, such as people, vehicles, or animals, may be a trigger condition and a direction of motion that will take the object out of the current camera's field of view may be another trigger condition. If the trigger conditions are met, trigger logic **216** may access graph map **218** and path logic **220** to determine which other camera or cameras should be modified and, in some configurations, what those modifications should be. For example, graph map **218** may be represented as a reference table or data structure **250** and may include entries **258** for each other video camera identifier **252**, child node identifiers **254**, and/or child node locations **256**, such as coordinates for the coordinate positioning system. In some configurations, near edge entries or fields for each child node identifier **254** and/or location **256**

may indicate a start of an object path that may be used to determine, from the event direction and using path logic **220**, shared child nodes for reaching one or more other cameras. For example, if a current field of view, such as field of view **206.1**, and the position and/or movement of object of interest **240** indicate detected near edge **210.10** for object of interest **240**, camera **110.1** may determine shared child nodes **208.2**, **208.1**, and **208.3** in graph map **218** for cameras **110.2** and **110.3** from reference table **250**. Path logic **220** may include a set of logical rules for determining predicted object paths based on graph map **218** and/or reference table **250**. In some configurations, camera **110.1** may select cameras **110.2** and **110.3** to receive alert messages or similar video capture update messages based on shared child node identifiers in the event direction.

Cameras **110** may include a communication channel **222** for sending one or more messages, such as a message to update video capture parameters to prepare for an incoming object, to other cameras. For example, cameras **110** may use a camera identifier **252** determined from reference table **250** for the most likely camera(s) to next see the object and include a messaging protocol to address a video capture update message to that camera identifier. In some configurations, cameras **110** may be internet protocol (IP) cameras on a common network and communication channel **222** may be configured for peer-to-peer communications through a network interface. In some configurations, cameras **110** may include a network or direct connection to a host and communication channel **222** may be configured to only communicate with the host. In such configurations, the host may be configured as a router for communications among cameras and/or be configured to process the message from the detecting camera and generate a corresponding video capture update message to send to the predicted camera(s).

Cameras **110** may include adjustment logic **224** configured to receive video capture update messages and adjust video capture operating parameters based on such messages. For example, adjustment logic **224** may receive and parse the message from the detecting camera to determine the shared child node identifier and be able to determine the changes in video capture operating parameters corresponding to an object approaching from the shared child node. In some configurations, a set of video capture operating parameters based on the object type, predicted direction of entry into the field of view, and/or other parameters to assist in modifying the video capture of the predicted camera may be determined based on the received message and shared child node identifier. In some configurations, reference table **250** may be configured with sets of operating parameters indexed by object type, child node identifier, and the applicable set of operating parameters may be included in the message or the object parameters in the message may allow the receiving camera to determine the set of operating parameters from the table. Adjustment logic **224** may use the modified parameters to initiate changes in the operation of camera actuators, encoders, analysis, and/or storage functions in response to the message. In some configurations, adjustment logic **224** in the receiving camera may also be configured to recognize when that camera is already engaged in high-priority video capture (such as due to a different object of interest or event) and may reject predictive updates that would negatively impact the current video capture operations.

Cameras **110** may include non-volatile memory **230** configured to store video data captured by that video camera. As described above, on-camera memory may include a limited capacity and cameras **110** may send video data through

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communication channel **222** to be stored in other memory systems, which may include the memory of other cameras, a host, and/or video storage accessible through the host. In some configurations, modification of video capture operating parameters may include increasing the quality of captured video (in terms of both sample rate and encoding) and a commensurate increase in the amount of storage capacity used to store such video.

FIG. **3** schematically shows selected modules of a surveillance system **300** configured for predictively modifying the video capture operations of selected cameras based on a graph map and a shared child node. Surveillance system **300** may incorporate elements and configurations similar to those shown in FIGS. **1-2**. For example, surveillance system **300** may be configured in a network video recorder similar to network video recorder **130**. In some embodiments, one or more of the selected modules may access or be instantiated in the processors, memories, and other resources of video cameras configured for video capture, similar to video cameras **110**, and/or user devices configured for video monitoring, similar to user device **170**. For example, a video camera and its embedded or attached compute resources may be configured with some or all functions of surveillance controller **330** and/or those functions may be shared between the video camera controller and a network video recorder or video surveillance as a service (VSaaS) server. Similarly, some or all of analytics engine **340** may be instantiated in the video camera and/or shared with other surveillance system components.

Surveillance system **300** and/or any component thereof may include a bus **310** interconnecting at least one processor **312**, at least one memory **314**, and at least one interface, such as video camera interface **316** and network interface **318**. Bus **310** may include one or more conductors that permit communication among the components of surveillance system **300**. Processor **312** may include any type of processor or microprocessor that interprets and executes instructions or operations. Memory **314** may include a random access memory (RAM) or another type of dynamic storage device that stores information and instructions for execution by processor **312** and/or a read only memory (ROM) or another type of static storage device that stores static information and instructions for use by processor **312** and/or any suitable storage element such as a hard disk or a solid state storage element.

Video camera interface **316** may be configured for connection with one or more video cameras. For example, video camera interface **316** may include a plurality of ethernet ports and supporting protocols compatible with PoE standards for connecting to a plurality of cameras. In some embodiments, video camera interface **316** may include a PoE network switch for providing power to connected cameras and routing data packets to and from connected cameras, such as control and video data.

Network interface **318** may include one or more wired or wireless network connections to network, similar to network **102**. Network interface **318** may include a physical interface, such as an ethernet port, and related hardware and software protocols for communication over the network, such as a network interface card or wireless adapter.

Surveillance system **300** may include one or more non-volatile memory devices **320** configured to store video data. For example, non-volatile memory devices **320** may include a plurality of flash memory packages organized as an addressable memory array and/or one or more solid state drives or hard disk drives. In some embodiments, non-volatile memory devices **320** may include a plurality of

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storage devices within, attached to, or accessible by video cameras and/or a network video recorder for storing and accessing video data.

Surveillance system **300** may include a plurality of modules or subsystems that are stored and/or instantiated in memory **314** for execution by processor **312** as instructions or operations. For example, memory **314** may include a surveillance controller **330** configured to control a plurality of video cameras, capture and store video streams from those cameras, and enable user access, such as through surveillance application **350**. Memory **314** may include an analytics engine configured to analyze video data to detect events for use by surveillance controller **330** and/or surveillance application **350**. Memory **314** may include a surveillance application configured to provide a user interface for monitoring, reviewing, and managing surveillance video and/or surveillance system **300**.

Surveillance controller **330** may include interface protocols, functions, parameters, and data structures for connecting to and controlling cameras, capturing and storing video data from those cameras, and interfacing with analytics engine **340** and surveillance application **350**. For example, surveillance controller **330** may be an embedded firmware application and corresponding hardware in a network video recorder configured for network and/or direct communication with a set of associated video cameras. Surveillance controller **330** may be configured as a central collection point for video streams from the associated video cameras that enables analysis of captured video data by analytics engine **340** and presentation of video streams and video event alerts to a user through surveillance application **350**. In some embodiments, some or all functions of surveillance controller **330** may be onboard each video camera and no network video recorder may be required.

In some embodiments, surveillance controller **330** may include a plurality of hardware and/or software modules configured to use processor **312** and memory **314** to handle or manage defined operations of surveillance controller **330**. For example, surveillance controller **330** may include a camera control interface **332**, a video capture interface **334**, a video storage interface **336**, and an access and display manager **338**.

Camera control interface **332** may include camera interface protocols and a set of functions, parameters, and data structures for using, configuring, communicating with, and providing command messages to cameras through video camera interface **316** and/or network interface **318**. For example, camera control interface **332** may include an API and command set for interacting with control circuit in each camera to access one or more camera functions. In some embodiments, camera control interface **332** may be configured to set video configuration parameters for camera image sensors, microphones, and/or video encoders, access pan-tilt-zoom features, set or modify camera-based motion detection, tripwire, object detection, and/or low light detection parameters, and/or otherwise manage operation of cameras. For example, camera control interface **332** may maintain a video camera configuration table, pages, or similar data structures that includes entries for each video camera being managed and their respective camera-specific configuration parameters, active control features (such as PTZ control), and other configuration and control information for managing the cameras. In some embodiments, each camera may be assigned a unique camera identifier that may be used by surveillance controller **330**, analytics engine **340**, and/or surveillance application **350** to associate video data with the camera from which it was received.

In some embodiments, camera control interface **332** may include a message interface **332.1** for sending and/or receiving control messages with one or more cameras. For example, message interface **332.1** may include a messaging protocol for adjusting operating parameter values for each camera. In some configurations, camera-specific configuration parameters may include a range of operating values and corresponding conditions for using those operating parameters for video capture operations. For example, different sets of configuration parameters may be used for different times, environmental conditions, or operating modes for each camera. Message interface **332.1** may enable surveillance controller **330** to modify the current operating configuration parameters of a camera. For example, message interface **332.1** may be configured to send a video capture update message to change one or more video capture parameters for a selected video camera. In some configurations, message interface **332.1** may use internet protocol, master-slave, and/or multi-master messaging over video camera interface **316** and/or network interface **318** to send messages to selected cameras. In some configurations, application-level messaging may be used to send, parse, and respond to messages and enable the cameras to determine parameter changes from the message content to initiate changes in video capture operations. In some configurations, message interface **332.1** may enable surveillance controller **330** to directly change memory locations in the cameras that store the configuration parameters to be changed using one or more update messages. In some configurations, message interface **332.1** may be integrated in or responsive to video capture interface **334** to enable surveillance controller **330** to respond to video events by sending video capture update messages to selected cameras.

In some embodiments, camera control interface **332** may include a PTZ controller **332.2** for one or more video cameras. For example, each video camera may be equipped with a PTZ control unit and associated motor and/or digital controllers and command interface for moving that video camera from its current position to pan, zoom, and/or tilt to change the field of view. In some embodiments, PTZ controller **332.2** may include a remote controller unit that sends PTZ control commands to adjust camera position and/or zoom in real-time, such as responsive to the detection of an object of interest in the field of view (but not ideally positioned in the field of view). In some embodiments, PTZ controller **332.2** may include a set of configuration settings for auto-tracking or auto-capture functions within the selected video camera. For example, one or more cameras may include an auto-capture feature for detecting objects of interest and then centering and zooming on the detected object. PTZ controller **332.2** may be used to configure the parameters of the auto-capture feature, such as classes of objects (e.g., humans, human faces, vehicles, license plates, etc.) to be captured, PTZ ranges or limits, timing, quality or reliability thresholds, etc. In some embodiments, PTZ controller **332.2** may be configured for responding to an update message (from message interface **332.1**) regarding video events that are outside the current field of view of the camera in order to predictively change video capture operating parameters in anticipation of an object entering the field of view. For example, responsive to a video event for an object approaching from the direction of another camera, PTZ controller **332.2** may be commanded to pan the camera toward the other camera or an anticipated intercept point of an object of interest in the adjusted field of view.

Video capture interface **334** may include camera interface protocols and a set of functions, parameters, and data

structures for receiving video streams from associated cameras. For example, video capture interface **334** may include video data channels and related data buffers for managing a plurality of camera video data streams. In some embodiments, each video camera may be allocated a dedicated video channel for continuously and/or selectively sending its video stream to video capture interface **334**. For example, each video camera configured as a primary video camera may have a dedicated video channel for their respective primary video streams. Video capture interface **334** may be configured to pass each received video stream to video storage interface **336**, analytics engine **340**, and/or access/display manager **338**. For example, received video streams may be buffered by video capture interface before being streamed to video storage interface **336**, analytics engine **340**, and access/display manager **338**. In some embodiments, video capture interface **334** may receive camera video metadata describing the camera video format, time and location information, and event or condition tags based on on-board camera analytics. Video capture interface **334** may generate additional video metadata for video format changes and provide video metadata to video storage interface **336** and/or other components. In some embodiments, video capture interface **334** may support an audio channel or audio track for audio data synchronized to the captured video data. For example, one or more video formats supported may include one or more audio channels for audio data from one or more microphones associated with the video camera. In some embodiments, video capture interface **334** may use video events detected from one camera to trigger changes in video capture operations, such as video capture operating modes and video capture rates, of one or more other video cameras in surveillance system **300**.

In some embodiments, video capture interface **334** may include a video stream manager **334.1** configured to identify and manage the plurality of video streams being received from the cameras. For example, video stream manager **334.1** may manage video buffer allocation and space, processing of video streams from a camera video format to another video format, flushing of buffered video to storage through video storage interface **336** and/or for display via access/display manager **338**. In some embodiments, video stream manager **334.1** may send video streams to analytics engine **340** for analysis and/or provide a notification to analytics engine **340** of the availability and storage location of video data for analysis in non-volatile memory **320** (as determined by video storage interface **336**). In some embodiments, video stream manager **334.1** may include configurable video paths. For example, the storage path (through video storage interface **336**), the display path (through access/display manager **338**), and/or the analytics path (through analytics engine **340**) may each be configured for specific processing, priority, and timing. In some embodiments, one or more selectable storage paths and corresponding storage locations or storage modes may be associated with different video capture operating modes and/or video capture rates and may be selected in response to triggers, such as video event triggers.

In some embodiments, video stream manager **334.1** may be configured to use encoder/decoder **334.2** to encode camera video streams in a desired video format. In some embodiments, encoder/decoder **334.2** may be configured to receive the raw video data stream from the image sensor and determine the video data format to be used, including a capture rate of the raw data from the image sensor. For example, the video capture rate may include a number of pixels or resolution of the image data from the image sensor and a frame rate determining how frequently the pixel values

are determined. In some embodiments, encoder/decoder 334.2 may support two or more selectable video capture rates and corresponding video formats. For example, encoder/decoder 334.2 may support a passive video capture rate 334.3 and an active video capture rate 334.4. Passive video capture rate 334.3 may have a lower rate than active video capture rate 334.4 such that passive video capture rate 334.3 captures less video data (e.g., low quality video 320.3) and requires less processing, network bandwidth, and/or data storage, but with lower video quality that may be less effective for display and/or analytics. Active video capture rate 334.4 may capture more video data with higher video quality (e.g., high quality video 320.3) for display and analytics, but at the cost of increased processor use, network bandwidth, and/or data storage. In some embodiments, video encoder/decoder 334.2 may support more than two selectable video capture rates. In some embodiments, in-camera video encoders may encode video data from the image sensor in a first (camera) video format and video stream manager 334.1 may use encoder/decoder 334.2 to re-encode them in one or more other formats. For example, video stream manager 334.1 may use encoder/decoder 334.2 to change the resolution, image size, frame rate, codec, compression factor, color/gray-scale, or other video format parameters.

In some embodiments, video capture interface 334 may include a plurality of operating modes 334.5. For example, a standby operating mode may include a low power state in which the video camera is not actively capturing video and a normal operating mode may include a normal power state where the video camera may activate any of its resources, including capturing video using its image sensor and processor. In some embodiments, operating modes 334.5 may include low light operating modes for low light conditions, motion/tripwire-only modes where only low-level processing of image sensor data (or a separate motion sensor) is active without video capture, and other operating modes. In some embodiments, one or more operating modes may be associated with specific video capture rates and encoding formats. For example, a passive video capture mode may use passive video capture rate 334.3 and an active video capture mode may use active video capture rate 334.4. Video cameras may use different operating modes during different operating periods and operating modes may be used to modify one or more video capture operations. For example, each operating mode may define whether or not video capture is suspended or operating and, if operating, what set of video capture parameters are being used, such as video capture rate (resolution and frame rate), encoding codec, filters, etc.

In some embodiments, video capture interface 334 may include trigger conditions 334.6 for moving between operating modes 334.5. For example, video capture interface 334 may have a default operating mode and one or more other operating modes that are triggered when specific conditions are met. For example, a camera may be configured to default to normal operating mode but change to low-light operating mode when light levels drop below a threshold. In some configurations, a camera may include passive video capture mode as a default mode for continuous video surveillance, due to the lower resource usage, and include one or more triggers conditions 334.6 for changing the operating mode. For example, when motion, a video tripwire, or an object are detected from the passive video data, the video capture interface 334 may change the operating mode to the active video capture mode for the camera that detected the condition. However, these video-based triggers may be less reli-

able using passive video data and there may be a lag between an object entering the field of view and successfully triggering a video condition, including the required processing time for the monitoring analytics. In some embodiments, video capture interface 334 may include one or more trigger conditions 334.6 based on predictively changing operating modes based on video events occurring at other video cameras. For example, video capture interface 334 may include a video event detector 334.7 that determines video events from each camera and uses graph map 334.8 and/or reference table 334.12 to determine which other cameras should change operating modes and what those operating mode changes should be.

Video event detector 334.7 may support detection of one or more video event types based on the video data received by surveillance controller 330. For example, incoming video streams for each camera may be processed through one or more analytical models, such as object detector 334.18, to determine whether an object of interest appears within a frame of video data at a selected confidence threshold. In some configurations, video event detector 334.7 may use video event parameters generated from one or more sources, such as video metadata, time, camera-detected events (e.g., motion sensors, video tripwires, low-weight on-board object detectors, etc.), and output from one or more analytics engines. For example, an object may be detected by a camera's onboard object detector and, based on the type of object detector, assigned an object type parameter, as well as detection time, and location in frame. In some configurations, an actuator position (and corresponding field of view) may be mapped to an edge in graph map 334.8 to determine an event direction. For example, the actuator position may define a set of vectors defining the field of view and an object position in the field of view may correspond to a path represented by an edge in graph map 334.8 that may be referred to as an event detected edge. In some configurations, an analytics engine with an object motion tracking model may then process a series of video frames in the video data stream to determine a direction of travel of the detected object. The object detection may trigger video event detector 334.7 to identify a video event and initiate event response logic 334.17 to evaluate graph map 334.8 and/or reference table 334.12 to determine whether video capture parameters of one or more other cameras should be updated. In some configurations, video event detector 334.7 may isolate video frames or a portion of the video data (e.g., bounding box) corresponding to an object of interest to include among the video event parameters, along with other object data.

Graph map 334.8 may include a data structure for representing the surveillance environment and video cameras as a set of nodes and edges, such as described above with regard to FIG. 2. For example, graph map 334.8 may include a set of parent nodes corresponding to the video cameras in the surveillance system and a set of child nodes mapped to object travel paths among the video cameras and, more specifically, the intersections of those paths. In some configurations, graph map 334.8 may include a plurality of graph maps configured for different object and/or event types 334.9. For example, one graph map may be configured for pedestrian traffic and model the walking paths among the locations where the cameras are located and another graph map may be configured for vehicular traffic and model the driving paths among the locations. Similarly, different graph maps may be configured for different times of day or conditions. For example, some paths may be closed during certain times of day and this may be reflected in different sets of child nodes and edges.

Graph map **334.8** may be based on a coordinate position system comprised of a coordinate grid and scale **334.10**. For example, latitude and longitude coordinates may be used as the coordinate grid. Alternatively, another grid system may be laid out for a campus, building complex, city, or other area of interest based on relevant features to define an alternate coordinate grid. Coordinate position systems may include two-dimensional coordinate layouts, such as longitude and latitude or another X-Y coordinate system. In some configurations, a three-dimensional coordinate layout may be used, such as adding elevation, altitude, or a similar Z coordinate to the two-dimensional coordinate position system. Based on the coordinate position system each camera and interconnecting paths may be modeled in graph map **334.8** as nodes and edges **334.11**. Using the coordinate position system, each camera may be defined as a parent node with a specific coordinate position (e.g., X-Y values) and each intersection may be defined as a child node with its own coordinate position. Edges may be defined as lines interconnecting the parent nodes and child nodes. In some configurations, graph map **334.8** may be represented as a data structure, such as a data table or array, comprised of coordinates for each node and pairs of nodes defining edges. In some configurations, edge data may enable calculation of corresponding path lengths for each edge and the path lengths for multiple edges traversed for a path between and across nodes may be determined from aggregating those path lengths.

Reference table **334.12** may include a data structure for using video event parameters to determine which other cameras would benefit from a change in operating mode to improve the likelihood of capturing additional video data relevant to the triggering video event. For example, where an object of interest is detected by one camera, reference table **334.12** may include entries for each other camera that indicates their spatial relationship to the detecting camera based on graph map **334.8**. In some configurations, reference table **334.12** may include a video event index **334.13** comprising one or more index fields corresponding to relevant event parameters. For example, video event index **334.13** may include a combination of type of object and event direction parameters for a detected object of interest. An event detected edge determined from camera and object positions and graph map **334.8** may be used as an index value for video event index **334.13**. In some configurations, each camera entry **334.14** in reference table **334.12** may include a camera identifier and location parameters that define the field of view, parent node coordinates, and child nodes for that camera. For example, the same camera identifier used for managing camera configuration parameters may be used and the location parameters may include parent node coordinates, child node coordinates, and possible object detection edges based on the actuator/field of view ranges of that camera. In some configurations, each entry may also include a set of video capture parameters **334.15** that correspond to an increased likelihood of capturing a subsequent video event in the field of view of that camera. For example, video capture parameters **334.15** for a spatially adjacent camera may include active capture rate parameters and PTZ control parameters for moving the camera to the nearest position for its field of view and a predicted edge to a shared child node. In some configurations, reference table **334.12** may include multiple entries for each camera relative to each other camera and indexed by event direction and/or detected event edge parameters to determine whether and which other cameras identifiers and

sets of video capture parameters are selected in response to any given video event (based on shared child nodes in the event direction).

In some embodiments, event response logic **334.16** may include logical rules for applying video events and their corresponding video event parameters to trigger conditions **334.6** for triggering one or more operating modes **334.5**. For example, video response logic **334.16** may include a set of threshold values and logical rules for applying those threshold values to triggering modification of the video capture operations. In some configurations, video response logic **334.16** may include a set of active capture thresholds for initiating active video capture rate **334.3** and/or a corresponding active video capture mode. In some configurations, event response logic **334.16** may use graph map **334.8** and/or reference table **334.12** for determine whether and what cameras should have their video capture parameters changed. For example, video event detector **334.7** may use video event parameters to search video event index **334.13** to determine whether the video event should trigger operating mode changes for any of the other video cameras. If no video event index values match the corresponding video event parameters, then no update messages may be initiated. Similarly, if multiple entries are returned in the search for different cameras, then update messages may be sent to multiple cameras. In some configurations, event response logic **334.16** may use graph map **334.8** and/or reference table **334.12** to determine a set of child nodes relative to the detecting camera that are shared with other cameras. The set of shared child nodes may be filtered based on the event direction and/or object detected edge to include only the shared child nodes in that direction. Event response logic **334.16** may generate a video capture update message to send to the cameras with shared child nodes based on the identified child node. For example, message interface **332.1** may be used to send a child node alert **334.17** that includes a child node identifier for the shared child node and that shared child node identifier may be used by the receiving camera to determine the video capture parameter changes to be made, such as adjusting the camera position toward the shared child node.

In some configurations, threshold values used by event response logic **334.16** may include confidence thresholds for the object detection or classification and/or direction of motion. For example, video event detector **334.7** may return a video event with an object type, type confidence value, direction of motion value, and motion confidence value. If the object type matches an object of interest type, the type confidence value meets a type confidence threshold (e.g., 50%), and the direction of motion value meets the motion confidence threshold (e.g., 50%), then video response logic **334.16** may trigger message interface **332.1** to send a video capture update message to the selected camera or cameras. In some embodiments, additional and/or alternative logical rule sets and/or entries in reference table **334.12** for applying video event parameters to initializing video capture, triggering PTZ movement toward a predicted intercept location with the camera field of view, and/or changing data storage and/or processing paths may be included in event response logic **334.16**.

In some embodiments, video capture interface **334** may be configured with an object detector **334.18** supporting detection of one or more object classes, such as people, animals, motor vehicles, etc. For example, object detector **334.18** may operate on captured video data received from a video camera to detect whether an object of interest is present in the video data. In some embodiments, object

detector **334.18** may include a lightweight object detection model that can be processed in near real-time with the limited processing bandwidth of the video camera and/or associated compute resources. In some embodiments, the object detection model may operate on the video data in the video stream and return a flag or class for the object type detected, an object quality of confidence metric, object position, and/or object boundary data, such as two horizontal positions and two vertical positions to define a boundary box within the video frame. In some embodiments, object detector **334.18** may have one or more associated object confidence thresholds for evaluating the object confidence value for each object detection event. For example, object detector **334.18** may include an object detection threshold, below which the presence of an object is not treated as sufficiently certain to raise an object detection event, such as 50% reliability. In some embodiments, object detector **334.18** may be used to identify object detection events and raise corresponding alerts to video event detector **334.16**.

In some embodiments, video capture interface **334** may include a predictive capture timer **334.19** configured to determine how long a camera that predictively changes its video capture operations should wait to see whether the object of interest is detected or return to default operation. For example, predictive capture timer **334.19** may use a predefined operating period after the sending of the update message and/or change of parameters by the selected camera to determine a duration to wait for the object of interest to be detectable within its field of view. In some configurations, the duration for the predictive operating period may be a default value, predetermined for each spatial relationship (e.g., provided in the corresponding entry in reference table **334.12**), and/or dynamically generated based on the distance between cameras and a velocity from the object motion model. In some configurations, an object path length may be calculated from the edges traversed between the two camera parent nodes, where each edge between nodes traversed for the path are added together to determine the path length. If multiple shared child nodes and/or paths exist between the parent nodes, then multiple path lengths may be determined and used, such as by averaging or using the maximum or minimum path length. Once the duration for the predictive operating period is determined for a particular video event and camera, predictive capture timer **334.19** may monitor the elapsed time and return the camera to its default operating mode when the duration is met. For example, when the time has elapsed, message interface **332.1** may send another update message to return the camera to default operation. Alternatively, the parameters sent in the original update message may include a duration parameter and the camera may execute its own predictive capture timer to automatically return to default operation when the time has elapsed.

Video storage interface **336** may include storage interface protocols and a set of functions, parameters, and data structures for managing storage of video data **320.1** in non-volatile memory **320**, such as storage devices and/or network video storage, for later retrieval and use by access/display manager **338** and/or analytics engine **340**. For example, video storage interface **336** may write camera video stream data from video data buffers and/or storage path video data from video capture interface **334** to non-volatile memory **320**. In some embodiments, audio data may be stored in separate audio data files or objects and/or as synchronized audio data in audio tracks or channels of video data **320.1**. In some embodiments, video storage interface **336** may include a storage manager **336.1** configured to manage video storage space in non-volatile memory **320** in

accordance with one or more operating modes **334.5**, data retention and/or data archiving schemes. For example, surveillance system **300** may support continuous and/or triggered recording of video data from associated cameras and storage manager **336.1** may include logic for enforcing a data retention and overwriting policy whereby the fixed storage space of non-volatile memory **320** is recycled for storing a recent period of captured video, video data meeting specific retention criteria, and/or deleting or archiving video data after one or more periods of time defined in the data retention policy. In some embodiments, storage manager **336.1** may support different storage locations for high quality video **320.2** captured during active video capture modes and low quality video **320.3** captured during passive video capture modes. Video storage interface **336** may also include a metadata manager **336.2** to receive and store video metadata as tags or metadata tracks in the video data or in an associated metadata table, file, or similar data structure associated with the corresponding video data objects.

Access/display manager **338** may include APIs and a set of functions, parameters, and data structures for displaying video from video capture interface **334** and/or video storage interface **336** to a user display application, such as surveillance application **350**. For example, access/display manager **338** may include a monitoring or display configuration for displaying one or more video streams in real-time or near real-time on a graphical user display of a user device and/or receive video navigation commands from the user device to selectively display stored video data from non-volatile memory **320**. In some embodiments, access/display manager **338** may maintain an index of real-time/near real-time video streams and/or stored or archived video streams that are available for access by surveillance application **350**. In some embodiments, the video index may include a corresponding metadata index that includes video data parameters (e.g., time, location, camera identifier, format, low light/normal light, etc.), detected audio and video event metadata (event time, location, type, parameters, etc.), and/or video management parameters (expiration, active/archive, access control, etc.) for use in displaying and managing video data. Access/display manager **338** may be configured to support surveillance application **350** when instantiated in the same computing device as surveillance controller **330**, directly attached to the computing device hosting surveillance controller **330**, and/or via a network within a LAN, WAN, VPN, or the internet. In some embodiments, access/display manager **338** may provide selective access to user video format **336.2** and/or video streams allocated by video capture interface **334** to the user video path.

Analytics engine **340** may include interface protocols, functions, parameters, and data structures for analyzing video data to detect video events, add them to video metadata, and/or raise alerts, such as through surveillance application **350**. For example, analytics engine **340** may be an embedded firmware application and corresponding hardware in a network video recorder configured for local analysis of video data captured from associated video cameras and may be integral to or accessible by surveillance controller **330**. In some embodiments, analytics engine **340** may run on a separate computing device from surveillance controller **330**, such as a video camera with analytics capabilities, a dedicated analytics appliance, data storage system with analytics capabilities, or a cloud-based analytics service. In some embodiments, analytics engine **340** may operate in real-time or near real-time on video data received by video capture interface **334**, delayed processing of video data stored by video storage interface **336**, and/or a combi-

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nation thereof based on the nature (and processing requirements) of the video events, volume of video to be processed, and other factors. In some embodiments, surveillance system 200 may comprise a plurality of analytics engines configured for a particular type of event and corresponding event detection algorithm or model.

In some embodiments, analytics engine 340 may include a plurality of hardware and/or software modules configured to use processor 312 and memory 314 to handle or manage defined operations of analytics engine 340. For example, analytics engine 340 may include an event manager 342 and analytical model library 344. Analytics engine 340 may be configured to run one or more event detection algorithms for determining, tagging, and/or initiating alerts or other actions in response to detected video events. In some embodiments, analytics engine 340 may be configured to tag or build metadata structures that map detected events to time and image location markers for the video stream from which they are detected. For example, analytics engine 340 may use motion, tripwire, object recognition, facial recognition, audio detection, speech recognition, and/or other algorithms to determine events occurring in a video stream and tag them in a corresponding metadata track and/or separate metadata table associated with the video data object.

Event manager 342 may include storage interface and/or buffer interface protocols and a set of functions, parameters, and data structures for processing target video streams for predefined event types and updating or adding metadata parameters describing the detected video events. For example, event manager 342 may be configured to process all incoming video streams for surveillance controller 330 and/or selectively process video data based on user selections (through surveillance application 350) or metadata criteria received from the video camera or video capture interface 334. In some embodiments, event manager 342 may include, support, or supplement event detection by surveillance controller 330, such as video events from video event detector 334.7 and/or object detection events from object detector 334.18. In some embodiments, event manager 342 may include a video selector 342.1 configured to select target video streams or video data sets for analysis, including associated audio data. For example, video selector 342.1 may identify a real-time video stream for near real time analysis or a bounded video data set, such as video with a specified camera identifier and timestamps between beginning and ending time markers and/or including a defined set of metadata parameters. Event manager 342 may include an event type selector configured to determine one or more event types to be detected from the selected video data. For example, an analytics configuration may be configured to analyze the selected video stream for a predefined set of audio event detection, motion detection, tripwire detection, object recognition, facial recognition, speech recognition, and/or similar video event types. Each event type may map or correspond to an analytical model type, set of parameters, and one or model weights for defining an event detection algorithm stored in analytical model library 344 for use by analytics engine 340 to detect potential video events (and/or predictive audio events).

Analytical model library 344 may include an API and set of functions, parameters, and data structures for storing a plurality of analytical models for use by analytics engine 340 in processing video data. For example, analytical model library 344 may include a plurality of trained analytical models and corresponding event detection algorithms for different event types, target object classes (e.g., cars, license plates, equipment, people, etc.), and/or training conditions.

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In some embodiments, analytical model library 344 may also support audio analytical models and/or combined video with audio analytical models. Each analytical model may include a set of base equation(s) for the analytical model type, set of target parameters, and one or model weights that determine the event detection algorithm that will be used for the event detection processing. In some embodiments, at least some analytical models may be machine learning based models trained based on one or more sets of relevant reference data. For example, the base equations may be trained using the reference data sets to determine the model weights to be used in the resulting analytical model. Trained analytical models may be deployed in analytics engine 340 and/or surveillance controller 330. In some embodiments, analytics engine 340 may include or access training services for generating (training) or updating (retraining) analytical models in analytical model library 344.

In some embodiments, analytical model library 344 may include at least one object recognition model 344.1. For example, a motor vehicle recognition model may apply a set of weighted parameter values representing the relationships among sets of feature vectors for comparison with reference data (such as sets of feature vectors for known motor vehicles) and determination of a probabilistic reliability or correlation factor. Analytical model library 344 may include or access object reference data 344.2 for matching detected objects with previously identified (or recognized) reference objects. For example, the motor vehicle recognition model may be applied to a reference database of relevant motor vehicle images and/or feature sets extracted therefrom to provide vehicle reference data. In some embodiments, for any given detected object, object recognition model 344.1 may return one or more recognized matches and a corresponding reliability value. For example, the motor vehicle recognition model may return one or more known individuals from the reference data and corresponding reliability values, assuming at least one match is found that meets a threshold reliability value. In some configurations, the video event parameters for an initial object detection event may be used as reference data for matching a similar object detected on other video data from cameras. For example, once an object of interest is detected, the bounding box and image data it contains may be designated as reference data for subsequent object recognition calculations of other cameras detecting an object of the same type in response to receiving an update message.

In some embodiments, analytical model library 344 may include at least one object motion model 344.3 for tracking and predicting movement of an object of interest based on video data. For example, object motion model 344.3 may include an object motion tracking model that detects the same object frame-to-frame. In some configurations, object motion model 344.3 may leverage the bounding boxes for an object from frame-to-frame to track the motion of that object through the video frame. The motion of the detected object in the frame may then be translated by a spatial model into a likely direction of travel for the object itself. For example, for a particular camera position, movement to the right in frame may roughly correlate to northward movement in the surveillance environment and left movement may correlate to southward movement. Changes in size of the object may also be used to estimate movement into or out of the frame. In some configurations, object motion model 344.3 may be used to determine a direction of movement parameter for video event detector 334.7. Other models for determining or approximating direction of travel of an object of interest within video data may also be employed.

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Surveillance application 350 may include interface protocols, functions, parameters, and data structures for providing a user interface for monitoring and reviewing surveillance video and/or managing surveillance system 300, such as through surveillance controller 330. For example, surveillance application 350 may be a software application running on a user device integral to, connected to, or in network communication with surveillance controller 330 and/or a hosting network video recorder. In some embodiments, surveillance application 350 may run on a separate computing device from surveillance controller 330, such as a personal computer, mobile device, or other user device. In some embodiments, surveillance application 350 may be configured to interact with APIs presented by access/display manager 338.

In some embodiments, surveillance application 350 may include a plurality of hardware and/or software modules configured to use processor 312 and memory 314 to handle or manage defined operations of surveillance application 350. For example, surveillance application 350 may include a video manager, an alert manager, and an analytics manager.

The video manager may include APIs and a set of functions, parameters, and data structures for navigating and displaying video streams from video cameras accessed through surveillance controller 330. For example, the video manager may include a graphical user interface and interactive controls for displaying lists, tables, thumbnails, or similar interface elements for selecting and displaying video streams for particular cameras, times, locations, and/or events. In some embodiments, the video manager may enable split screen display of multiple camera video streams. For example, the near real-time video streams (with a predetermined lag based on network lag, storage, and processing times) from all active cameras may be displayed on a monitoring interface or a set of video streams corresponding to a detected event may be displayed in an event review interface. In some embodiments, the video manager may include a data structure summarizing all video data stored in surveillance system 300 to enable the user to locate and view older surveillance video data. For example, a video management log or database may include entries for stored video data indexed by related metadata, such as video data parameters (e.g., time, location, camera identifier, format, low light/normal light, etc.), detected video event metadata (event time, location, type, parameters, etc.), and/or video management parameters (expiration, active/archive, access control, etc.).

The alert manager may include APIs and a set of functions, parameters, and data structures for setting, monitoring, and displaying alerts based on detected video events. For example, the user may define a set of trigger events that generate visual, audible, tactile, and/or notification-based (electronic mail, text message, automated call, etc.) alerts to a user device. In some embodiments, the alert manager may include a plurality of preset alert conditions with associated event parameters and allow a user to enable and disable alert types and/or change associated event parameters. In some embodiments, the alert manager may be configured to operate in conjunction with an event overlay function to overlay graphical elements representing detected events or event indicators on video streams displayed through the video manager. For example, detected motion, objects, or faces may be boxed or highlighted, tagged with relevant identifiers, or otherwise indicated in the video playback on the user device.

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The analytics manager may include APIs and a set of functions, parameters, and data structures for selecting, training, and managing event detection algorithms. For example, the analytics manager may include a user interface to analytical model library 344 for one or more analytics engines 340. In some embodiments, the event detection algorithms may include a set of parameters and/or model weights that are preconfigured based on training data sets processed independent of surveillance system 300. For example, the analytics manager may include audio event detection and object detection algorithms for common objects/sources, situations, and camera configurations. In some embodiments, the analytics manager may include access to training services and/or preconfigured training data sets. For example, the analytics manager may enable the user to define training data sets for determining or refining event detection algorithm parameters and/or model weights based on predefined base algorithms or models. In some embodiments, the analytics manager may interface directly with analytics engine 340 for selecting, training, managing, and using the event detection algorithms configured through the analytics manager. In some embodiments, the analytics manager may interface with access/display manager 338 for accessing and managing one or more analytics engines 340.

As shown in FIG. 4, surveillance system 300 may be operated according to an example method for sending an alert based on graph mapped relationships among video cameras, i.e., according to method 400 illustrated by blocks 410-418 in FIG. 4.

At block 410, video events may be monitored. For example, each video camera may capture video data and analyze it for objects of interest using an object detection model.

At block 412, whether an event has been detected may be determined. For example, one of the video cameras may detect an object of interest in its video data stream meeting the trigger conditions of its video event detector and method 500 may proceed to block 414. If no event is detected, method 400 may return to block 410 to continue to monitor for video events.

At block 414, child nodes on an event detected edge may be determined. For example, the detecting video camera may determine the event detected edge or event direction from the camera and/or object position and use the event detected edge to determine the child nodes connected to the event detected edge (based on the graph map or a reference table based on the graph map).

At block 416, parent nodes sharing child nodes on the event detected edge may be determined. For example, the set of child nodes determined at block 414 may be used to determine the parent nodes that share an edge connection to one or more of those child nodes.

At block 418, an alert message with the child node identifier may be sent to the parent nodes. For example, the detecting video camera may send a video capture update message including the child node identifier or identifiers for the shared child node or nodes in the event direction to each the video cameras corresponding to each parent node determined at block 416.

As shown in FIG. 5, surveillance system 300 may be operated according to an example method for modifying video capture by repositioning based on the child node in an alert, i.e., according to method 500 illustrated by blocks 510-520 in FIG. 5.

At block 510, alerts may be monitored. For example, each video camera may monitor for video capture update messages or similar alerts from other video cameras.

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At block **512**, whether an alert has been received may be determined. For example, one of the video cameras may detect an object of interest in its video data stream and send an alert according to method **400**. If an alert is received, method **500** may proceed to block **514**. If no alert is received, method **500** may return to block **510** to continue to monitor for alerts.

At block **514**, a child node identifier may be determined from the alert message. For example, the video capture update message may include the child node identifier as an alert parameter in the message and it may be parsed from the message by the receiving camera.

At block **516**, the child node location may be determined based on the graph map. For example, the child node identifier may include or correspond to a graph map or reference table entry that includes a coordinate position of that child node and the receiving camera may use one of these resources to determine the child node coordinates.

At block **518**, a target angle or field of view based on the child node location may be determined. For example, the receiving camera may be configured with actuator positions and fields of view corresponding to different paths of approach represented by edges connecting the parent node of the camera to different child nodes, and the camera may use the child node location to determine the target actuator position to use to face that child node location.

At block **520**, the camera may be repositioned to the target angle or field of view. For example, the receiving camera may adjust its actuator position to the target actuator position determined at block **518** to initiate video capture in case the object of interest enters its field of view from the direction of the child node intersection.

As shown in FIG. 6, surveillance system **300** may be operated according to an example method for predictive modification of video capture operations among networked video cameras based on a video event detected by one camera modifying the operation of another camera, i.e., according to method **600** illustrated by blocks **610-634** in FIG. 6.

At block **410**, a group of video cameras may be configured with respective fields of view and known spatial relationships between them represented in a graph map. For example, a user may deploy a set of networked video cameras in a surveillance location with fixed positions and fixed or moveable fields of view based on the orientation, size, and depth of field of the image sensors and associated lens(es) and map the cameras and paths for object movement among them on a graph map.

At block **612**, network communication may be established among the video cameras. For example, each camera may be configured with a network interface and corresponding network protocols for peer-to-peer communication and/or networked to a shared host, such as a network video recorder.

At block **614**, sets of video capture operating parameters may be determined. For example, each video camera may be configured to support two or more selectable video capture rates, such as a passive video capture rate and an active video capture rate, and/or other video capture operating parameters.

At block **616**, a default set of video capture operating parameters may be selected for each camera. For example, the video cameras may be configured to default to the passive video capture rate to conserve processing power, network bandwidth, and/or storage space until an object of interest is within or approaching the field of view.

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At block **618**, video data may be received from at least one video camera using a current set of video capture operating parameters. For example, a surveillance controller for the camera may continuously receive video data from the video sensor using the passive video capture rate.

At block **620**, a video event may be determined from the video data. For example, the surveillance controller for one of the video cameras may detect an object of interest in its field of view.

At block **622**, the video event direction of the video event may be determined. For example, the surveillance controller may determine, based on the current field of view and a corresponding edge aligned with the field of view, an event detected edge corresponding to the event direction.

At block **624**, another video camera may be determined based on the video event direction and graph map. For example, the surveillance controller may use the event detected edge to identify child nodes in the event direction and use the graph map to determine cameras that share the identified child nodes and would be most likely to have the object of interest enter their field of view.

At block **626**, a video capture update message with a child node identifier may be determined. For example, the surveillance controller may generate a message to the predicted camera (or cameras) to trigger a change in video capture operating parameters of that camera prior to the object of interest entering its field of view. The message may include the child node identifier for the shared child node to indicate the direction the object is likely to approach from.

At block **628**, the video capture update message may be sent to the predicted camera. For example, the surveillance controller may send or rout the generated message to the predicted camera based on a camera identifier and/or corresponding network address.

At block **630**, the other video camera may receive the video capture update message. For example, the predicted and selected camera may receive the message from the surveillance controller through its network interface.

At block **632**, video capture operations may be modified based on the video capture update message and child node identifier. For example, responsive to the message, the receiving video camera may extract the child node identifier and change one or more parameters to implement a different set of video capture operating parameters for capturing an object approaching from the direction of the child node.

At block **634**, video data may be captured by the other video camera based on the modified video capture operations. For example, following the modification of the set of video capture operating parameters, the receiving video camera may operate for some operating period using the new parameters in order to more reliably detect the object of interest if and when it enters its field of view.

As shown in FIG. 7, surveillance system **300** may be operated according to an example method for modifying video capture operations based on a video capture update message from an event detected by another camera, i.e., according to method **700** illustrated by blocks **710-742** in FIG. 7. In some embodiments, method **500** may operate in conjunction with one or more blocks of method **600** in FIG. 6.

At block **710**, video capture rates may be determined. For example, a video camera may be configured at manufacture or based on user settings with multiple video capture rates supporting different operating modes of the video camera.

At block **712**, operating modes may be determined. For example, the video camera may be configured at manufacture or based on user settings with multiple operating modes

that may include a default operating mode with a set of video capture operating parameters.

At block **714**, multiple camera positions may be determined. For example, upon installation, a video camera configured with PTZ functions enabled may determine ranges for pan, tilt, and/or zoom functions and the corresponding camera positions based on continuous and/or step adjustments and may align those camera positions with object paths corresponding to adjacent edges from the graph map.

At block **716**, storage modes may be determined. For example, upon system configuration, two or more storage locations for video data from the video camera may be determined, such as local storage in a flash memory of the video camera (in a first mode), storage to one or more disk drives in a network video recorder (in a second mode), and/or network video storage in a remote cloud video storage system (in a third mode).

At block **718**, a video capture rate may be selected. For example, upon initialization, the video camera may default to a video capture rate, such as a passive video capture rate.

At block **720**, a default object detection mode may be used. For example, upon initialization, the video camera may be configured to use a low-weight object detector operating on lower quality video data and/or a motion sensor or video trip wire until a possible video event is detected.

At block **722**, a camera position may be determined. For example, upon initialization or as determined by a user, the video camera may be placed in a default or last used position using the PTZ controls.

At block **724**, one or more initial storage modes may be determined. For example, upon initialization or based on default operating mode, the camera may default to a storage location, such as on-camera non-volatile memory.

At block **726**, an operating period may be initiated. For example, the camera may enter a default operating mode with an associated set of video capture operating parameters selected based on blocks **718-724**.

At block **728**, a video capture update message may be received. For example, based on another video camera detecting a video event for a moving object of interest and determining that this camera shared a child node in the graph map indicating a possible path for the object, this video camera may receive an alert message, including the child node identifier, to predictively change its video capture operating parameters to increase the likelihood of detecting the object of interest.

At block **730**, the child node identifier may be determined from the video capture update message. For example, the camera may parse the child node identifier from an alert parameter in the alert message.

At block **732**, a corresponding edge may be determined based on the child node identifier. For example, the camera may use the child node identifier to determine the corresponding edge from the graph map connecting the camera's parent node to the shared child node.

At block **734**, the actuator position for the corresponding edge and other video capture parameters may be selected. For example, the camera may use PZT control to move the field of view toward the edge corresponding to the likely direction of approach indicated by the child node. The camera may also adjust the video capture rate, operating modes, and/or storage modes to support capture of the object of interest.

At block **736**, a duration for the modified operating period may be determined. For example, the video camera may

receive or determine a duration to monitor for the object of interest and set a predictive capture timer to measure the elapsed time to the duration.

At block **738**, video data may be captured. For example, during the current operating period, the video camera may use the modified video capture operations to capture video data from the video image sensor.

At block **740**, video data may be stored. For example, during the current operating period, the video camera may send and store the captured video data for the modified video data operations.

At block **742**, the duration for modified video capture operations may be determined to have elapsed and the video camera may return to default video capture operating parameters. For example, if the predetermined duration is met without detecting the object of interest, the video camera may return to block **726** to initiate another operating period with the original set of video capture operating parameters.

As shown in FIG. **8**, surveillance system **300** may be operated according to an example method for configuring multiple network video cameras for use of graph maps for managing spatial relationships and alerts, i.e., according to method **800** illustrated by blocks **810-834** in FIG. **8**.

At block **810**, a set of video cameras are determined. For example, during installation of a video surveillance system, the installer may determine the number of video cameras being installed and their locations.

At block **812**, one or more object types may be determined for objects of interest. For example, the surveillance system may be configured for monitoring people, vehicles, animals, inventory, or another object of interest.

At block **814**, paths and intersections may be determined. For example, based on the surveillance system environment, the installer may map out the paths available for movement of the objects of interest and note where there are intersections or decision points regarding which path to follow.

At block **816**, a coordinate position system may be determined. For example, the installer may tag locations based on global positioning system (GPS) or longitude/latitude coordinates and/or implement another coordinate system for mapping the relative positions of the locations.

At block **818**, coordinates may be determined for the parent nodes. For example, each camera location may be configured as a parent node with the coordinates for that location.

At block **820**, coordinates may be determined for the child nodes. For example, each intersection may be configured as a child node with the coordinates for that location.

At block **822**, edges may be determined among nodes. For example, each pair of nodes that is connected by an object path may be represented as an edge between those nodes. The resulting graph map may represent the relative positions of the cameras and object paths for at least one object type.

At block **824**, a camera may be selected to configure. For example, each camera may be configured in turn to act as both a possible detecting camera and a predictively alerted camera using the graph map to adjust camera positioning and other video capture parameters. Each camera may include a reference table or similar data structure representing the entire graph map or a selected portion of the graph map related to that camera's relationships to other cameras that have shared child nodes.

At block **826**, shared child nodes and their coordinates may be determined. For example, for each other camera, the configuration may include the shared child nodes and their coordinates from the graph map.

At block **828**, child node entries may be stored. For example, the configuration may include entry with coordinates for each shared child node.

At block **830**, correlations between field of view positions and edges may be determined. For example, the camera may include views of multiple paths connected to different intersections and represented by different child nodes and connecting edges and the configuration may map those edges to the field of view and/or actuator positions for capturing those fields of view.

At block **832**, path distances and/or delay may be determined. For example, the configurations may include calculation of total path distances by one or more paths from each other camera through their shared child nodes to determine a total path distance that may be used to calculate delays and/or modified operating periods for waiting for an object of interest to arrive in the field of view.

At block **834**, a reference data structure may be stored for each video camera. For example, a parent node entry for each other video camera may be stored in the reference table of each video camera. Method **800** may return to block **824** to configure the entries for each camera and/or return to block **812** for generating multiple graph maps for different object types or conditions impacting the available paths.

As shown in FIG. 9, surveillance system **300** may be operated according to an example method for configuring and updating graph maps and reference data structures for expanding the set of video cameras, i.e., according to method **900** illustrated by blocks **910-930** in FIG. 9. In some embodiments, method **900** may operate in conjunction with one or more blocks of method **800** of FIG. 8.

At block **910**, a set of video cameras may be determined. For example, a surveillance system may be configured with an initial set of video cameras in a surveillance environment.

At block **912**, a graph map may be configured. For example, the graph map corresponding to the initial installation of the surveillance system may be configured using method **800**.

At block **914**, a reference data structure may be configured. For example, the graph map and related operating parameters for adjusting camera operation may be configured in a reference table or similar reference data structure.

At block **916**, the reference data structure may be deployed. For example, the reference table or a relevant portion of the table for a particular video camera may be stored in that video camera for use by its surveillance controller.

At block **918**, the set of video cameras may be operated using the reference data structure during an operating period. For example, the surveillance system may operate using the reference data structure to execute methods **400**, **500**, **600**, and/or **700** from FIGS. 4, 5, 6, and 7.

At block **920**, an additional set of video cameras may be determined. For example, the surveillance area may be expanded and new cameras may be added adjacent to the original surveillance area and cameras determined at block **910**.

At block **922**, new parent nodes may be added to the graph map. For example, the graph map may be expanded to include new parent nodes and coordinates for the new camera locations.

At block **924**, new child nodes may be added to the graph map. For example, the graph map may be expanded to include new paths among the new camera locations and the original camera locations and represent the new intersections as child nodes.

At block **926**, an updated set of edges may be determined for the graph map. For example, the graph map may be expanded to include edges among the new parent and child nodes, as well as adjacent nodes from the original graph map.

At block **928**, new parent node entries may be added to the reference data structure. For example, new parent node entries and coordinates for each new camera in the additional set of video cameras may be added to the reference data structure.

At block **930**, new child node entries may be added to the reference data structure. For example, new child node entries and coordinates for each new intersection may be added to the reference data structure. Method **900** may return to block **916** to deploy the updated reference data structure to the new video cameras and the original video cameras impacted by new shared child nodes.

While at least one exemplary embodiment has been presented in the foregoing detailed description of the technology, it should be appreciated that a vast number of variations may exist. It should also be appreciated that an exemplary embodiment or exemplary embodiments are examples, and are not intended to limit the scope, applicability, or configuration of the technology in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment of the technology, it being understood that various modifications may be made in a function and/or arrangement of elements described in an exemplary embodiment without departing from the scope of the technology, as set forth in the appended claims and their legal equivalents.

As will be appreciated by one of ordinary skill in the art, various aspects of the present technology may be embodied as a system, method, or computer program product. Accordingly, some aspects of the present technology may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.), or a combination of hardware and software aspects that may all generally be referred to herein as a circuit, module, system, and/or network. Furthermore, various aspects of the present technology may take the form of a computer program product embodied in one or more computer-readable mediums including computer-readable program code embodied thereon.

Any combination of one or more computer-readable mediums may be utilized. A computer-readable medium may be a computer-readable signal medium or a physical computer-readable storage medium. A physical computer-readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, crystal, polymer, electromagnetic, infrared, or semiconductor system, apparatus, or device, etc., or any suitable combination of the foregoing. Non-limiting examples of a physical computer-readable storage medium may include, but are not limited to, an electrical connection including one or more wires, a portable computer diskette, a hard disk, random access memory (RAM), read-only memory (ROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), a Flash memory, an optical fiber, a compact disk read-only memory (CD-ROM), an optical processor, a magnetic processor, etc., or any suitable combination of the foregoing. In the context of this document, a computer-readable storage medium may be any tangible medium that can contain or store a program or data for use by or in connection with an instruction execution system, apparatus, and/or device.

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Computer code embodied on a computer-readable medium may be transmitted using any appropriate medium, including but not limited to, wireless, wired, optical fiber cable, radio frequency (RF), etc., or any suitable combination of the foregoing. Computer code for carrying out operations for aspects of the present technology may be written in any static language, such as the C programming language or other similar programming language. The computer code may execute entirely on a user's computing device, partly on a user's computing device, as a stand-alone software package, partly on a user's computing device and partly on a remote computing device, or entirely on the remote computing device or a server. In the latter scenario, a remote computing device may be connected to a user's computing device through any type of network, or communication system, including, but not limited to, a local area network (LAN) or a wide area network (WAN), Converged Network, or the connection may be made to an external computer (e.g., through the Internet using an Internet Service Provider).

Various aspects of the present technology may be described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus, systems, and computer program products. It will be understood that each block of a flowchart illustration and/or a block diagram, and combinations of blocks in a flowchart illustration and/or block diagram, can be implemented by computer program instructions. These computer program instructions may be provided to a processing device (processor) of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which can execute via the processing device or other programmable data processing apparatus, create means for implementing the operations/acts specified in a flowchart and/or block(s) of a block diagram.

Some computer program instructions may also be stored in a computer-readable medium that can direct a computer, other programmable data processing apparatus, or other device(s) to operate in a particular manner, such that the instructions stored in a computer-readable medium to produce an article of manufacture including instructions that implement the operation/act specified in a flowchart and/or block(s) of a block diagram. Some computer program instructions may also be loaded onto a computing device, other programmable data processing apparatus, or other device(s) to cause a series of operational steps to be performed on the computing device, other programmable apparatus or other device(s) to produce a computer-implemented process such that the instructions executed by the computer or other programmable apparatus provide one or more processes for implementing the operation(s)/act(s) specified in a flowchart and/or block(s) of a block diagram.

A flowchart and/or block diagram in the above figures may illustrate an architecture, functionality, and/or operation of possible implementations of apparatus, systems, methods, and/or computer program products according to various aspects of the present technology. In this regard, a block in a flowchart or block diagram may represent a module, segment, or portion of code, which may comprise one or more executable instructions for implementing one or more specified logical functions. It should also be noted that, in some alternative aspects, some functions noted in a block may occur out of an order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or blocks may at times be executed in a reverse order, depending upon the operations involved. It will also be noted that a block of a block

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diagram and/or flowchart illustration or a combination of blocks in a block diagram and/or flowchart illustration, can be implemented by special purpose hardware-based systems that may perform one or more specified operations or acts, or combinations of special purpose hardware and computer instructions.

While one or more aspects of the present technology have been illustrated and discussed in detail, one of ordinary skill in the art will appreciate that modifications and/or adaptations to the various aspects may be made without departing from the scope of the present technology, as set forth in the following claims.

The invention claimed is:

1. A system, comprising:

a first video camera having a first field of view, wherein a plurality of networked video cameras includes the first video camera;

a controller in communication with the plurality of networked video cameras and configured to:

receive video data from the first video camera;

determine, from the video data, a first video event having an event direction;

determine, based on the event direction and a graph map of the plurality of networked video cameras, at least one other video camera from the plurality of networked video cameras that shares a child node in the graph map with first video camera, wherein the child node is in the event direction; and

send, to the at least one other video camera and based on the first video event, a video capture update message, wherein the video capture update message includes a child node identifier for the child node; and

a non-volatile memory configured to store a data structure based on the graph map and comprising:

a plurality of parent node entries for the plurality of networked video cameras;

a plurality of child node entries for a plurality of child nodes corresponding to path intersections between at least two video cameras of the plurality of video cameras; and

a set of coordinates, based on a coordinate position system of the graph map, for each child node entry of the plurality of child node entries, wherein the first video camera is further configured to use the data structure to determine, for the first video event, each shared child node in the event direction for the at least one other video camera.

2. The system of claim 1, further comprising:

the plurality of networked video cameras, wherein each video camera of the at least one other video camera is configured to, responsive to receiving the video capture update message:

determine the child node identifier from the video capture update message;

modify, based on the child node identifier, a video capture operation of that other video camera; and capture, using that other video camera, video data based on the modified video capture operation.

3. The system of claim 2, wherein:

the at least one other video camera includes an actuator configured to adjust a direction of a field of view of the at least one other video camera; and

modifying the video capture operation comprises:

operating, during a first operating period, in a first actuator position of a plurality of positions for adjusting the direction of the field of view; and

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selecting, responsive to the video capture update message and during a second operating period, a second actuator position of the plurality of positions.

4. The system of claim 3, wherein:
the graph map comprises:
a coordinate position system;
a first set of coordinates for the at least one other video camera; and
a second set of coordinates for the child node; and
the at least one other video camera is further configured to determine, based on the first set of coordinates and the second set of coordinates, the second actuator position.

5. The system of claim 2, wherein:
each video camera of the plurality of networked video cameras is configured for a plurality of video capture rates; and
modifying the video capture operation comprises:
operating, during a first operating period, at a first video capture rate from the plurality of video capture rates; and
selecting, responsive to the video capture update message and during a second operating period, a second video capture rate to modify the video capture operation.

6. The system of claim 2, wherein:
the graph map comprises:
a coordinate position system;
a first set of coordinates for the at least one other video camera;
a second set of coordinates for the child node; and
a third set of coordinates for the first video camera;
the at least one other video camera comprises a plurality of sets of video capture operating parameters;
the at least one other video camera is further configured to use a first set of video capture operating parameters during a first operating period; and
modifying the video capture operation comprises:
determining, based on the graph map, a path distance from the first video camera to the at least one other video camera through the child node;
determining, based on the path distance, a duration for a second operating period for determining, from video data of the at least one other video camera, a second video event related to the first video event; and
selecting, responsive to the video capture update message and during the second operating period, a second set of video capture operating parameters.

7. The system of claim 2, wherein:
the first video camera includes a plurality of graph maps;
determining the first video event comprises detecting, using an object detection model, an object of interest in the video data;
determining the at least one other video camera from the plurality of networked video cameras comprises:
determining a set of object parameters for the object of interest; and
using the set of object parameters for the object of interest to select the graph map from the plurality of graph maps; and
modifying the video capture operation of the at least one other video camera:
occurs prior to the object of interest entering a field of view of the at least one other video camera; and
increases a likelihood of detecting the object of interest.

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8. The system of claim 2, wherein the controller is further configured to:
determine, for each video camera of the plurality of networked video cameras, a reference data structure for that video camera, wherein each reference data structure comprises:
a set of parent node entries for parent nodes connected to that video camera through one or more child nodes in the graph map;
a set of shared child node entries for child nodes connected to that video camera in the graph map; and
a set of path distance parameters for each path through at least one child node to a parent node; and
send, to each video camera of the plurality of networked video cameras, the reference data structure for that video camera, wherein each video camera of the at least one other video cameras is further configured to store and use the reference data structure for that other video camera to modify the video capture operation for that other video camera.

9. The system of claim 1, wherein:
the data structure further comprises a field of view position for each edge connecting a parent node to at least one child node; and
the first video camera is further configured to:
determine, based on the first field of view, the event direction and a corresponding field of view position; and
determine, based on the corresponding field of view position, the child node.

10. The system of claim 1, wherein:
during a first operating period:
the plurality of networked video cameras comprises a first set of video cameras;
the graph map includes a first set of child nodes among the first set of video cameras; and
the data structure comprises a first configuration of parent node entries, child node entries, and sets of coordinates; and
during a second operating period:
the plurality of networked video cameras comprises the first set of video cameras and a second set of video cameras;
the graph map includes a second set of child nodes among the first set of video cameras and the second set of video cameras; and
the data structure comprises a second configuration comprising:
additional parent node entries for the second set of video cameras; and
additional child node entries and sets of coordinates for the second set of child nodes.

11. A computer-implemented method, comprising:
receiving video data from a first video camera of a plurality of networked video cameras;
determining, from the video data, a first video event having an event direction;
determining, based on the event direction and a graph map of the plurality of networked video cameras, at least one other video camera from the plurality of networked video cameras that shares a child node in the graph map with first video camera, wherein the child node is in the event direction;
sending, to the at least one other video camera and based on the first video event, a video capture update message, wherein the video capture update message includes a child node identifier for the child node;

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determining, by the at least one other video camera and responsive to receiving the video capture update message, the child node identifier from the video capture update message;

modifying, by the at least one other video camera and based on the child node identifier, a video capture operation of the at least one other video camera; and capturing, using the at least one other video camera, video data based on the modified video capture operation.

12. The computer-implemented method of claim 11, wherein:

the at least one other video camera includes an actuator configured to adjust a direction of a field of view of the at least one other video camera;

the graph map comprises:

a coordinate position system;

a first set of coordinates for the at least one other video camera; and

a second set of coordinates for the child node; and

modifying the video capture operation comprises:

operating, during a first operating period, in a first actuator position of a plurality of positions for adjusting the direction of the field of view;

determining, responsive to the video capture update message and based on the first set of coordinates and the second set of coordinates, a second actuator position; and

selecting, during a second operating period, the second actuator position of the plurality of positions.

13. The computer-implemented method of claim 11, wherein:

each video camera of the plurality of networked video cameras is configured for a plurality of video capture rates; and

modifying the video capture operation comprises:

operating, during a first operating period, at a first video capture rate from the plurality of video capture rates; and

selecting, responsive to the video capture update message and during a second operating period, a second video capture rate to modify the video capture operation.

14. The computer-implemented method of claim 11, wherein:

the graph map comprises:

a coordinate position system;

a first set of coordinates for the at least one other video camera;

a second set of coordinates for the child node; and

a third set of coordinates for the first video camera;

the at least one other video camera comprises a plurality of sets of video capture operating parameters; and

modifying the video capture operation comprises:

using a first set of video capture operating parameters during a first operating period;

determining, based on the graph map, a path distance from the first video camera to the at least one other video camera through the child node;

determining, based on the path distance, a duration for a second operating period for determining, from video data of the at least one other video camera, a second video event related to the first video event; and

selecting, responsive to the video capture update message and during the second operating period, a second set of video capture operating parameters.

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15. The computer-implemented method of claim 11, wherein:

the first video camera includes a plurality of graph maps; determining the first video event comprises detecting, using an object detection model, an object of interest in the video data;

determining the at least one other video camera from the plurality of networked video cameras comprises:

determining a set of object parameters for the object of interest; and

using the set of object parameters for the object of interest to select the graph map from the plurality of graph maps; and

modifying the video capture operation of the at least one other video camera:

occurs prior to the object of interest entering a field of view of the at least one other video camera; and

increases a likelihood of detecting the object of interest.

16. The computer-implemented method of claim 11, further comprising:

configuring, based on the graph map, a data structure for the first video camera comprising:

a plurality of parent node entries for the plurality of networked video cameras;

a plurality of child node entries for a plurality of child nodes corresponding to path intersections between at least two video cameras of the plurality of networked video cameras; and

a set of coordinates, based on a coordinate position system of the graph map, for each child node entry of the plurality of child node entries; and

determining, for the first video event and using the data structure, each shared child node in the event direction for the at least one other video camera.

17. The computer-implemented method of claim 16, further comprising:

determining, based on a field of view of the first video camera, the event direction and a corresponding field of view position, wherein the data structure further comprises a field of view position for each edge connecting a parent node to at least one child node; and

determining, based on the corresponding field of view position, the child node.

18. The computer-implemented method of claim 16, further comprising:

configuring, during a first operating period, the data structure in a first configurations of parent node entries, child node entries, and sets of coordinates, wherein:

the plurality of networked video cameras comprises a first set of video cameras during the first operating period; and

the graph map includes a first set of child nodes among the first set of video cameras in the first operating period; and

configuring, during a second operating period responsive to adding a second set of video cameras to the first set of video cameras in the plurality of networked video cameras, the data structure in a second configuration by:

adding, to the graph map, a second set of parent nodes corresponding to the second set of video cameras;

adding, to the graph map, a second set of child nodes among the first set of video cameras and the second set of video cameras;

determining, for the graph map, an updated set of edges among the parent nodes and the child nodes;

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adding, to the data structure, additional parent node entries for the second set of video cameras; and
 adding, to the data structure, additional child node entries and sets of coordinates for the second set of child nodes.

19. A system, comprising:
 a plurality of networked video cameras;
 a processor;
 a memory;
 means for receiving video data from a first video camera of a plurality of networked video cameras;
 means for determining, from the video data, a first video event having an event direction;
 means for determining, based on the event direction and a graph map of the plurality of networked video cameras, at least one other video camera from the plurality of networked video cameras that shares a child node in the graph map with first video camera, wherein the child node is in the event direction;
 means for sending, to the at least one other video camera and based on the first video event, a video capture update message, wherein the video capture update message includes a child node identifier for the child node;
 means for determining, by the at least one other video camera and responsive to receiving the video capture

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- update message, the child node identifier from the video capture update message;
 means for modifying, by the at least one other video camera and based on the child node identifier, a video capture operation of the at least one other video camera; and
 means for capturing, using the at least one other video camera, video data based on the modified video capture operation.
 20. The system of claim 19, further comprising:
 means for storing a data structure based on the graph map and for the first video camera, the data structure comprising:
 a plurality of parent node entries for the plurality of networked video cameras;
 a plurality of child node entries for a plurality of child nodes corresponding to path intersections between at least two video cameras of the plurality of networked video cameras; and
 a set of coordinates, based on a coordinate position system of the graph map, for each child node entry of the plurality of child node entries; and
 means for determining, for the first video event and using the data structure, each shared child node in the event direction for the at least one other video camera.

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